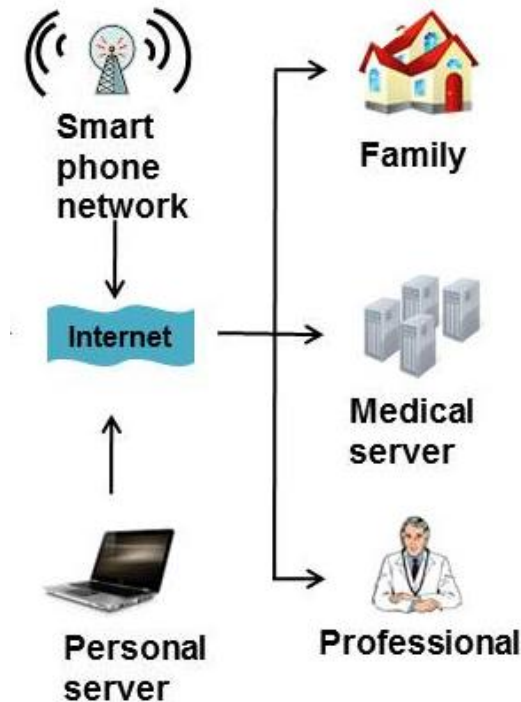
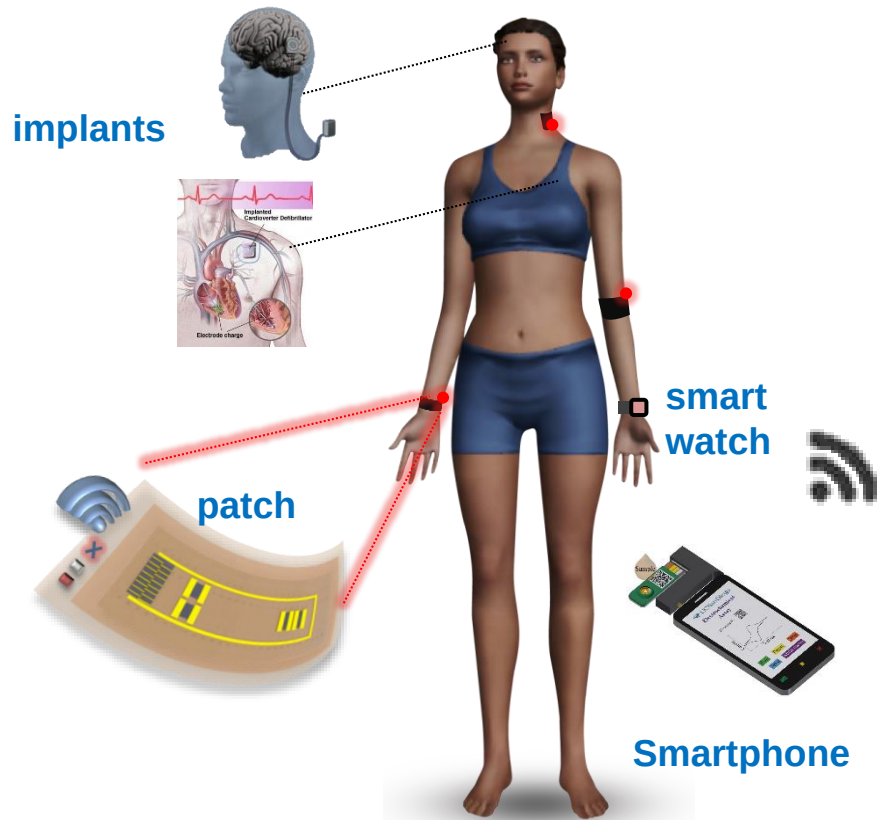


IOT in smart healthcare



- preventive medicine
- diagnostics
- prognostics
- rehabilitation
- therapy

IOT-enabled sensor systems which can monitor physiological and clinical parameters are promising for smart healthcare.

IOT-enabled sensor systems for smart healthcare



m
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T

Standalone



Smartphone-integrated

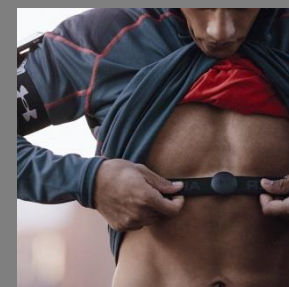


W
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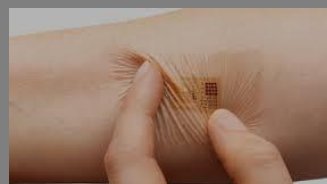
dressable



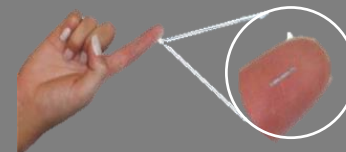
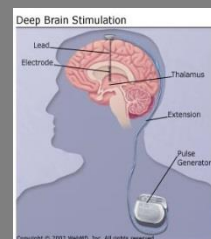
accessory



skin-attachable



implantable



U
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E

Mobile point-of-care testing (mPOCT) systems

Portable PoCTs : limits in connectivity and personalization



iSTAT
Abbott Labs



Blood analyzer
Alere



Accu-Check
Roche



Coagucheck
Roche



Stratus troponin
analyzer
Siemens

mPOCTs : advantage in connectivity (spatiotemporal mapping, epidemic demography, preventive healthcare) but limit of low accuracy and no standardization



Glucometer
iHealth



LFA reader
CELLMIC



SAW biosensor
OJbio



Heart monitor(ECG)
Alivecor

Fluorescence imaging-based high accuracy bioassay for mPOCT

Fluorescence
spectrophotometer



Agilent



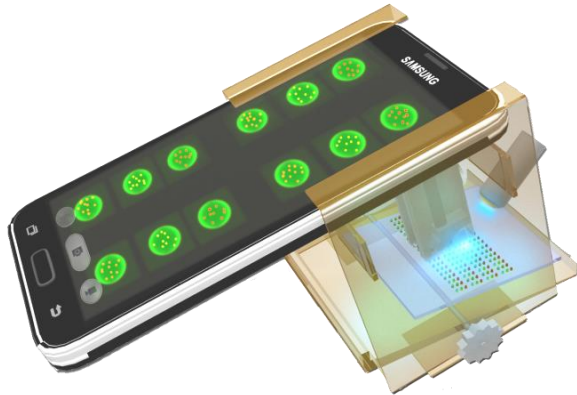
PMT

Hamamatsu



CMOS Image Sensor (CIS)

Rough exposure time control / low SNR



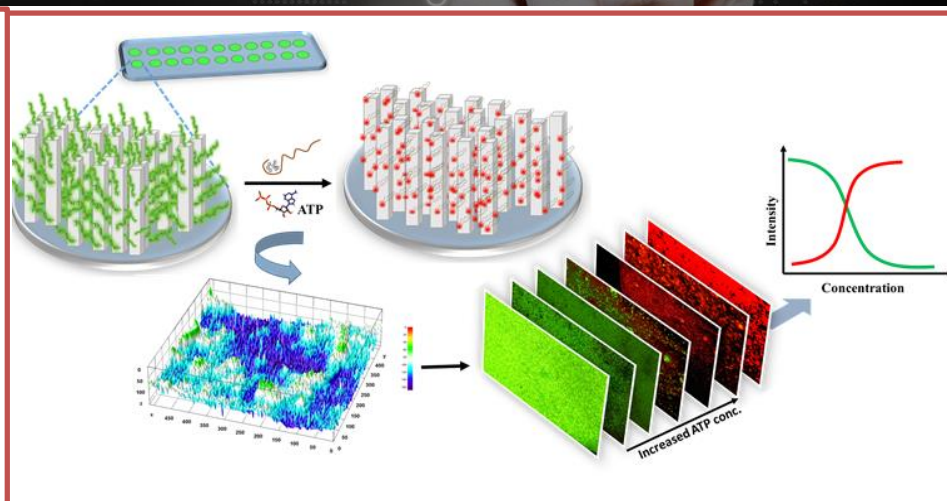
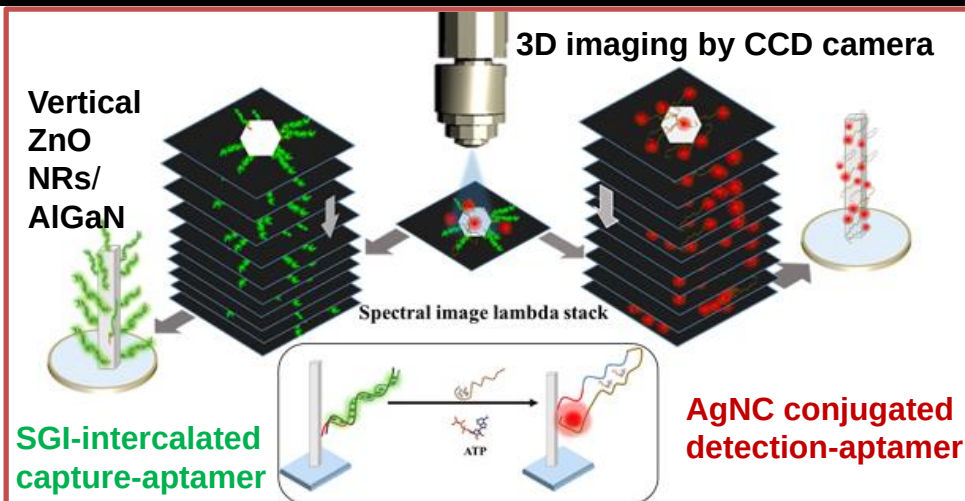
Smartphone

Issue : Low Accuracy

False-negatives/
False-positives ↓

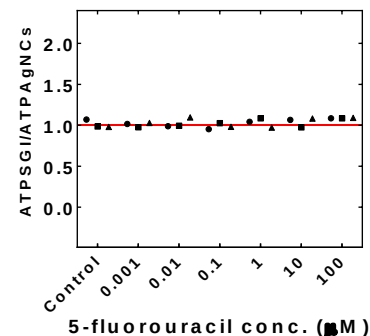
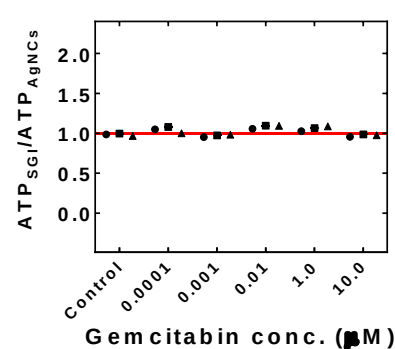
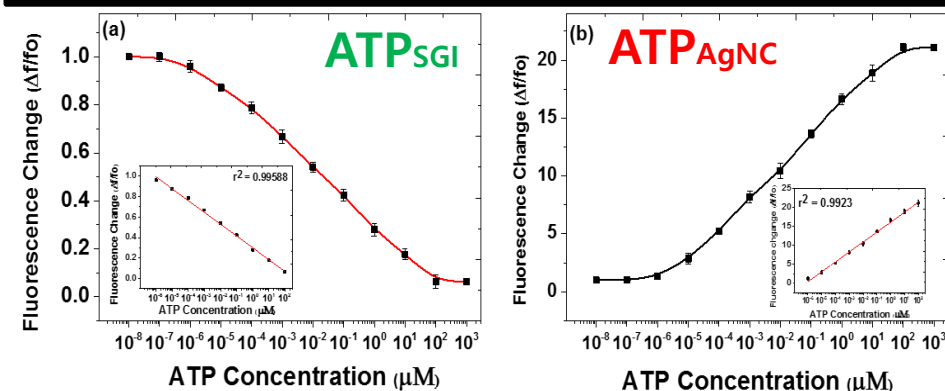
specificity & selectivity ↑

Fluorescence imaging-based 'seesawed' high-accuracy detection of biomolecules



ZnO NR-enhanced fluorescence (FRET),
Quantitative 3D quantification $\rightarrow$ SNR $\uparrow$

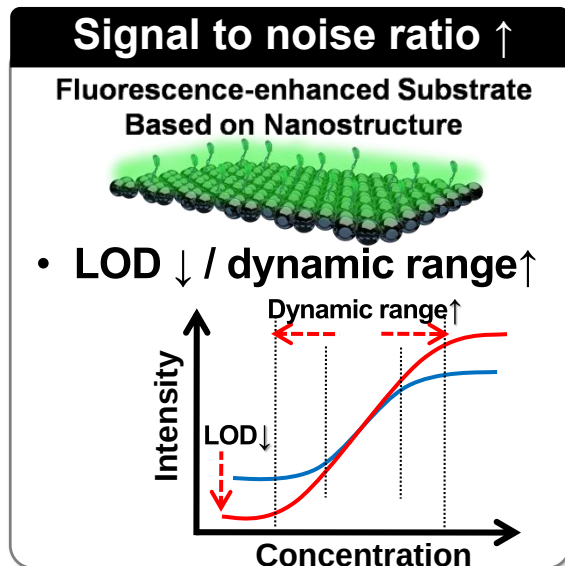
"Seesawed ratio" of red and green
fluorescence ~ 1 during cellular ATP



Minimize false-negatives : sensitivity $\uparrow$

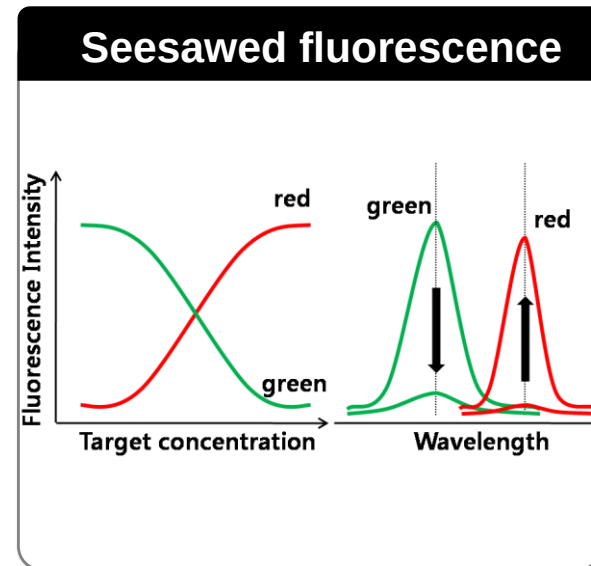
Minimize false-positives : selectivity $\uparrow$

Smartphone imaging-based fluorescence detection for high accuracy bioassays



False-negatives $\downarrow$

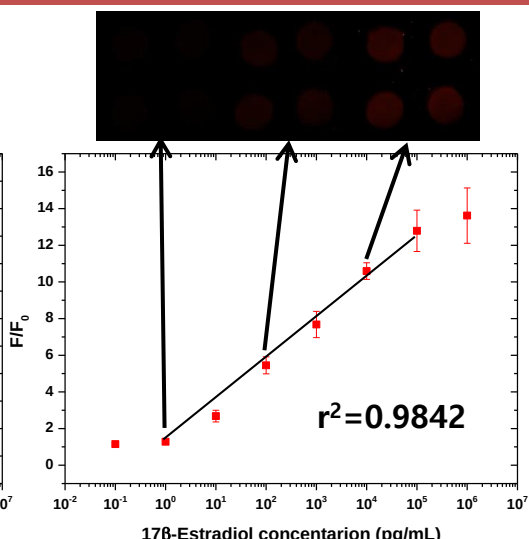
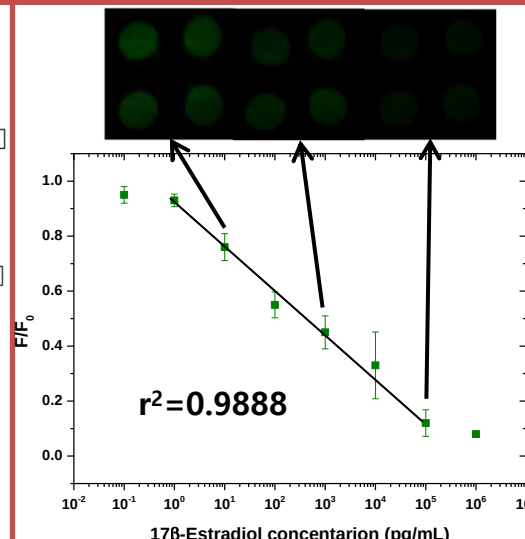
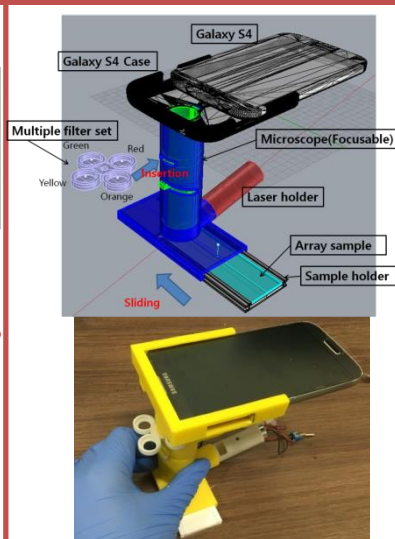
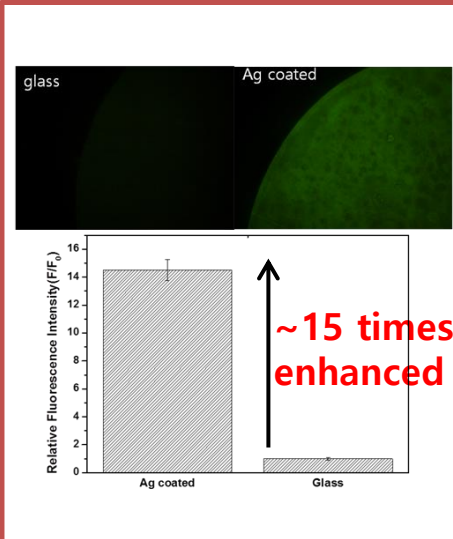
Sensitivity $\uparrow$



False-positives $\downarrow$

Selectivity $\uparrow$

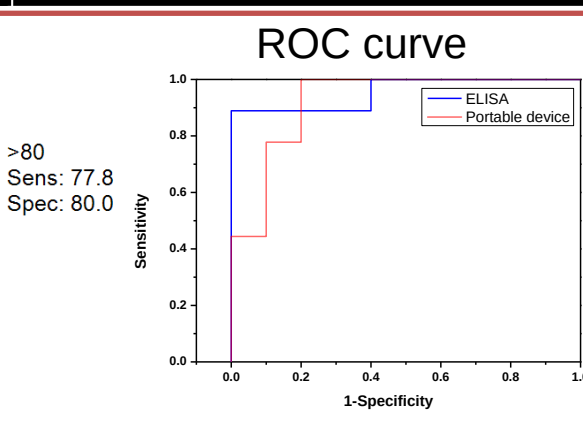
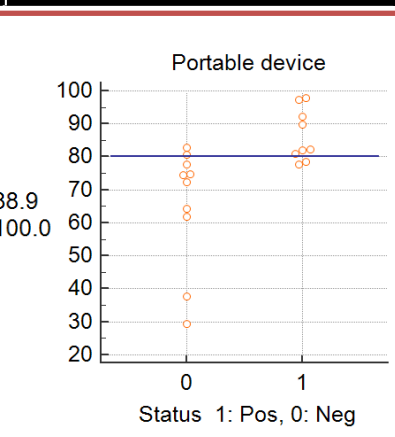
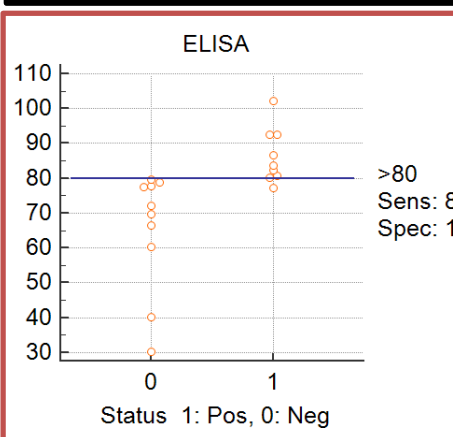
Smartphone imaging-based fluorescence detection for high accuracy bioassays



MEF(Ag) $\rightarrow$ SNR $\uparrow$

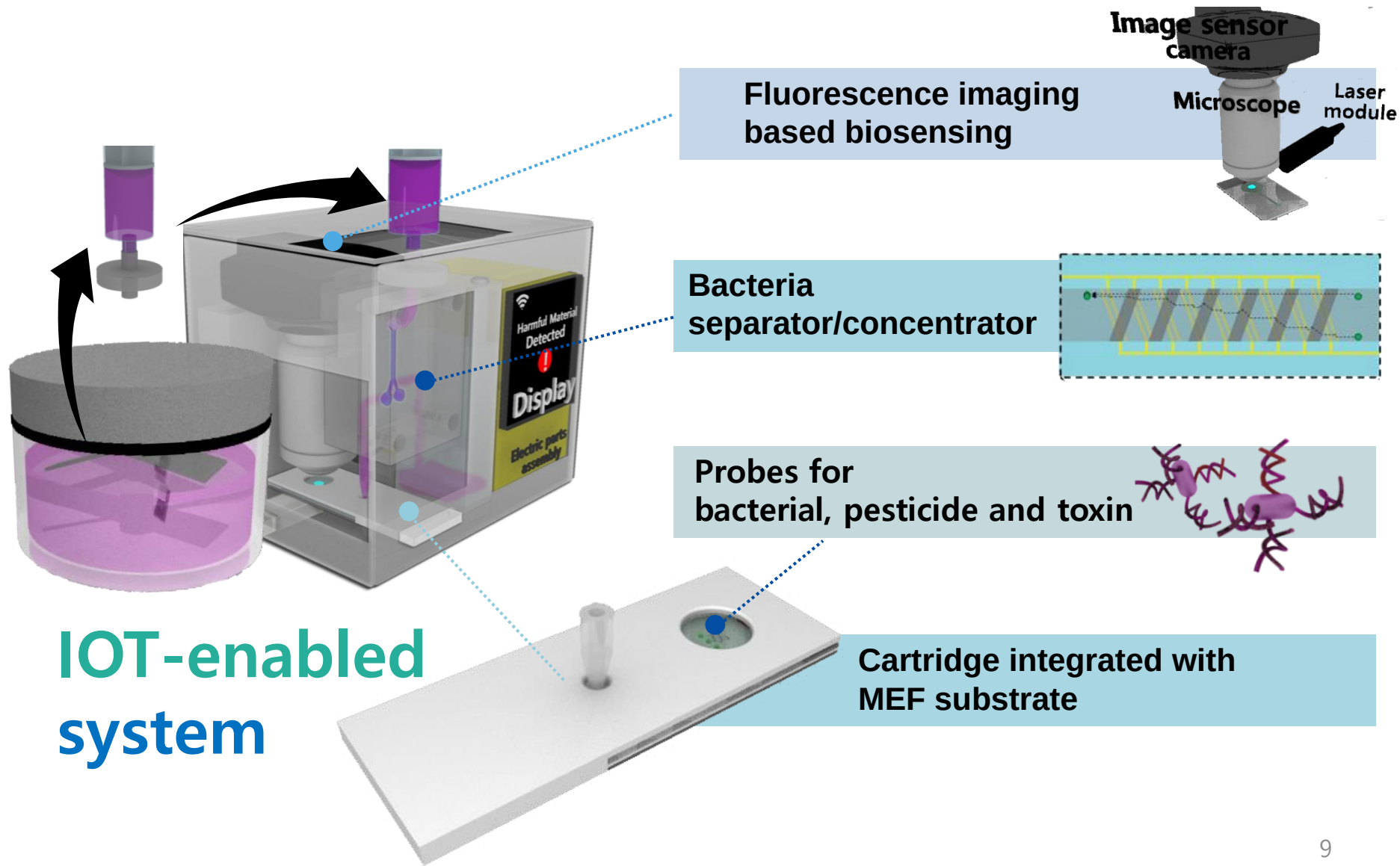
Prototyping of using 3D printing

Calibration curve for 17 β -estradiol with smartphone fluorescence microscope



Statistical accuracy test of target analyte spiked wastewater	
ELISA	Area under ROC curve
Mobile biosensor	0.956
	0.922

Fluorescence imaging-based on-the-spot detection system for food safety



Why skin-attachable sensor patches?

Accessory : non-invasive but **limit in unobtrusive** monitoring



Patch: non-invasive and **unobtrusive** monitoring,
high SNR due to conformal contact with skin



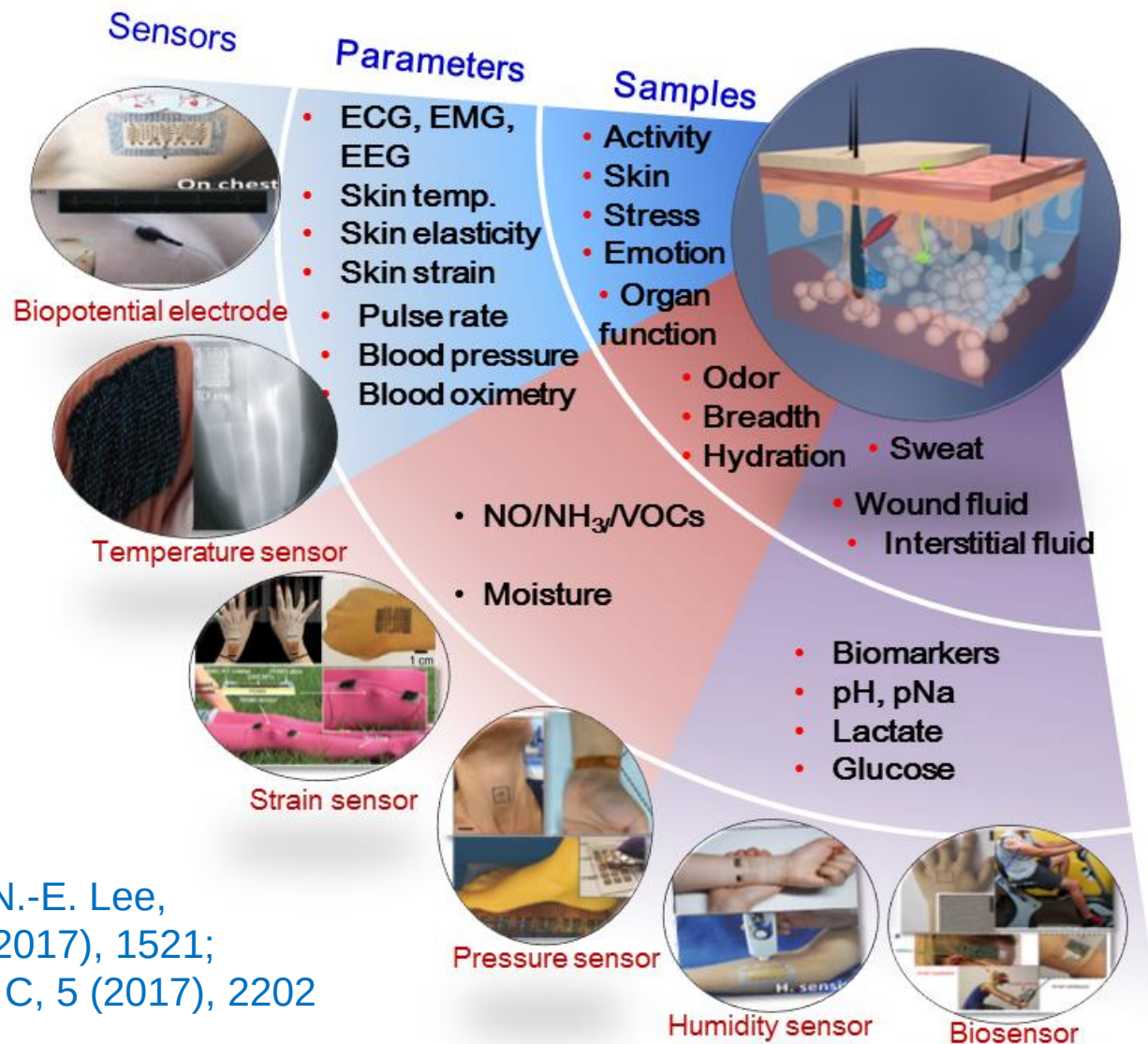
Biostamp
MC10



S-patch
Samsung



What can be measured by skin-attachable sensor patches?



T.Q. Trung and N.-E. Lee,
Adv. Mater. 29 (2017), 1521;
J. Mater. Chem. C, 5 (2017), 2202

n-attachable



Stretchable sensors for skin-attachable patches



Materials

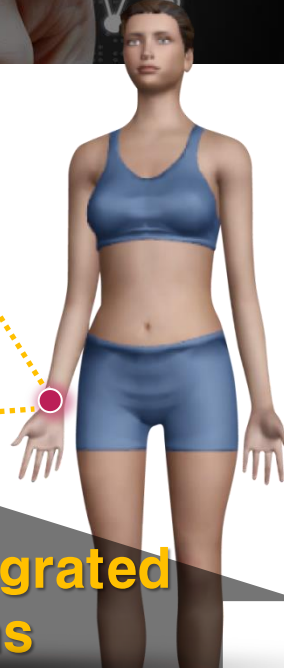
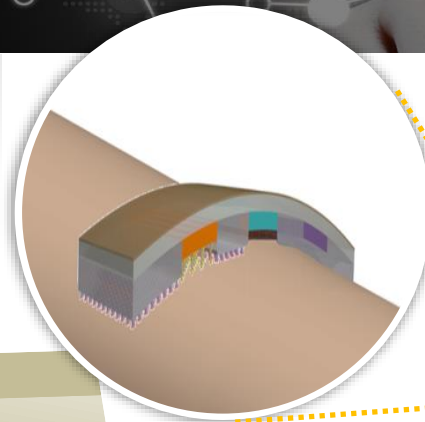
Stretchable sensing materials, electrochemical electrodes, dry biopotential electrodes

Device

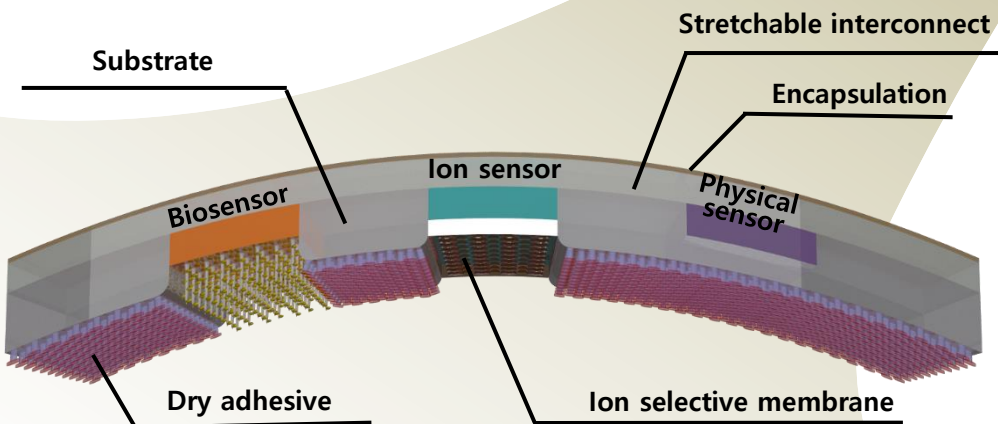
Stretchable sensors, energy harvesters, energy storage devices

Packaging

Substrate, dry adhesives, interconnect, encapsulation



Stretchable materials & devices



Sensor-integrated systems

Integration

Sensor array, integration of sensors, power, and MCU

S/W

Signal processing, data transmission, apps, big data

Clinical

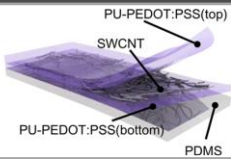
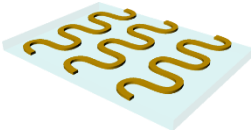
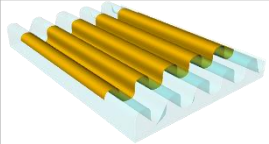
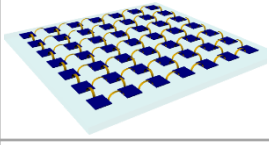
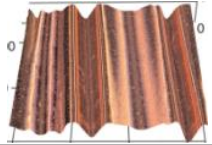
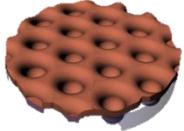
New applications, clinical evaluation, service

Stretchable physical sensors for skin-attachable patches



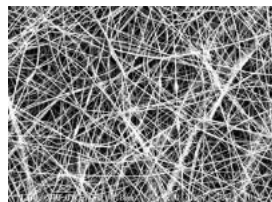
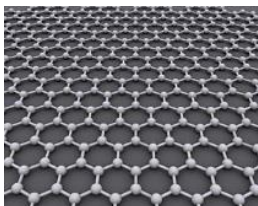
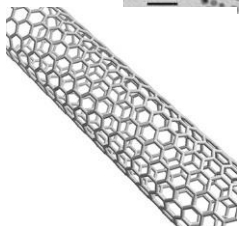
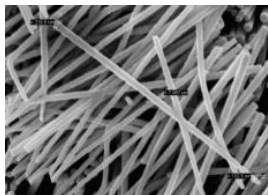
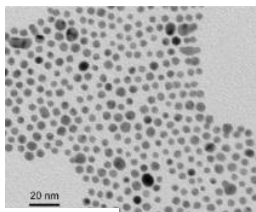
Approaches for stretchability



Materials	Strategies	Designs	Process methods	Stretchable direction
Intrinsically stretchable components	Using intrinsically stretchable materials	 Elastomeric nanocomposites	Spin-coating, printing, spraying <i>ACS Nano</i> 2015 , 9, 6252	Omni-direction
Geometric engineering of flexible materials	In-plane, geometric engineering	 Serpentine routing	Patterning <i>Appl. Phys. Lett.</i> 2014 , 104, 021908	Uniaxial
	Out-of-plane, geometric engineering	 Wavy structure	Pre-stretching and release <i>J. Vac. Sci. Technol. A</i> 2009 , 27, L9	Uniaxial, Biaxial
		 Island-bridge	Transfer printing <i>IEEE Trans. Compon. Packag. Manuf. Technol.</i> 2015 , PP, 1	Biaxial
		 Imperceptible	Transfer on pre-strained ultrathin substrate <i>Adv. Mater.</i> 2015 , 27, 34	Uniaxial
	Out-of-plane, 3D structuring	 Bio-mimicking	Soft lithography <i>Adv. Sci.</i> 2015 , 2, n/a	Multi-direction, but not fully stretchable
		 Microstructured pattern	Soft lithography spin coating, printing, spraying	Omni-direction

Approach 1: Intrinsically stretchable elastomeric nanocomposites

Nanomaterials



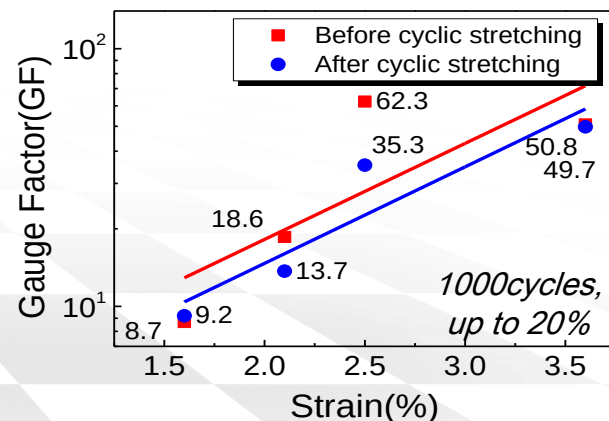
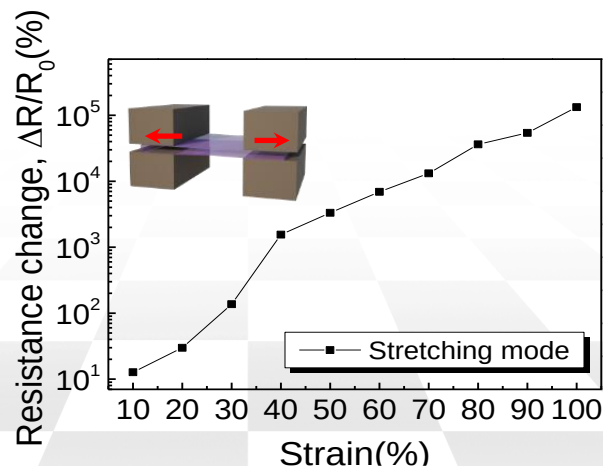
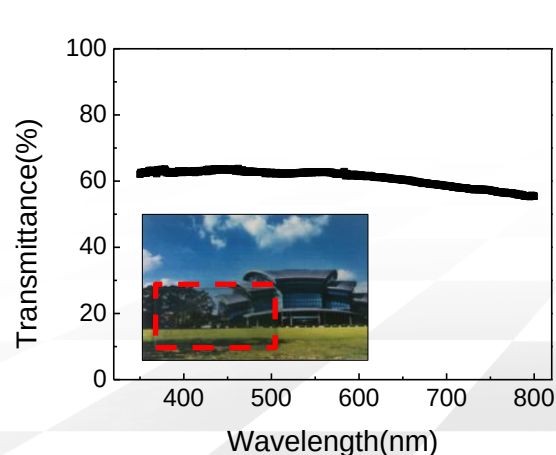
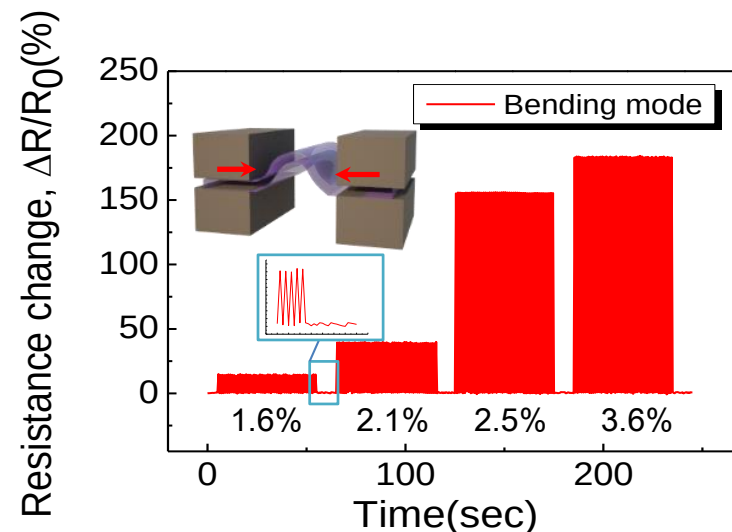
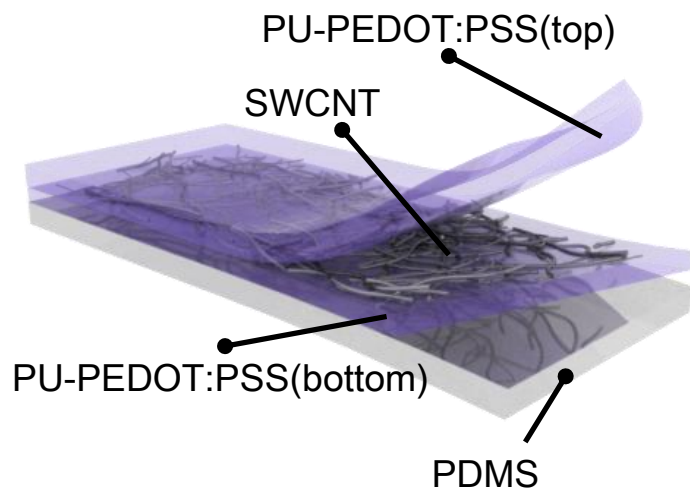
Elastomer

Transparent

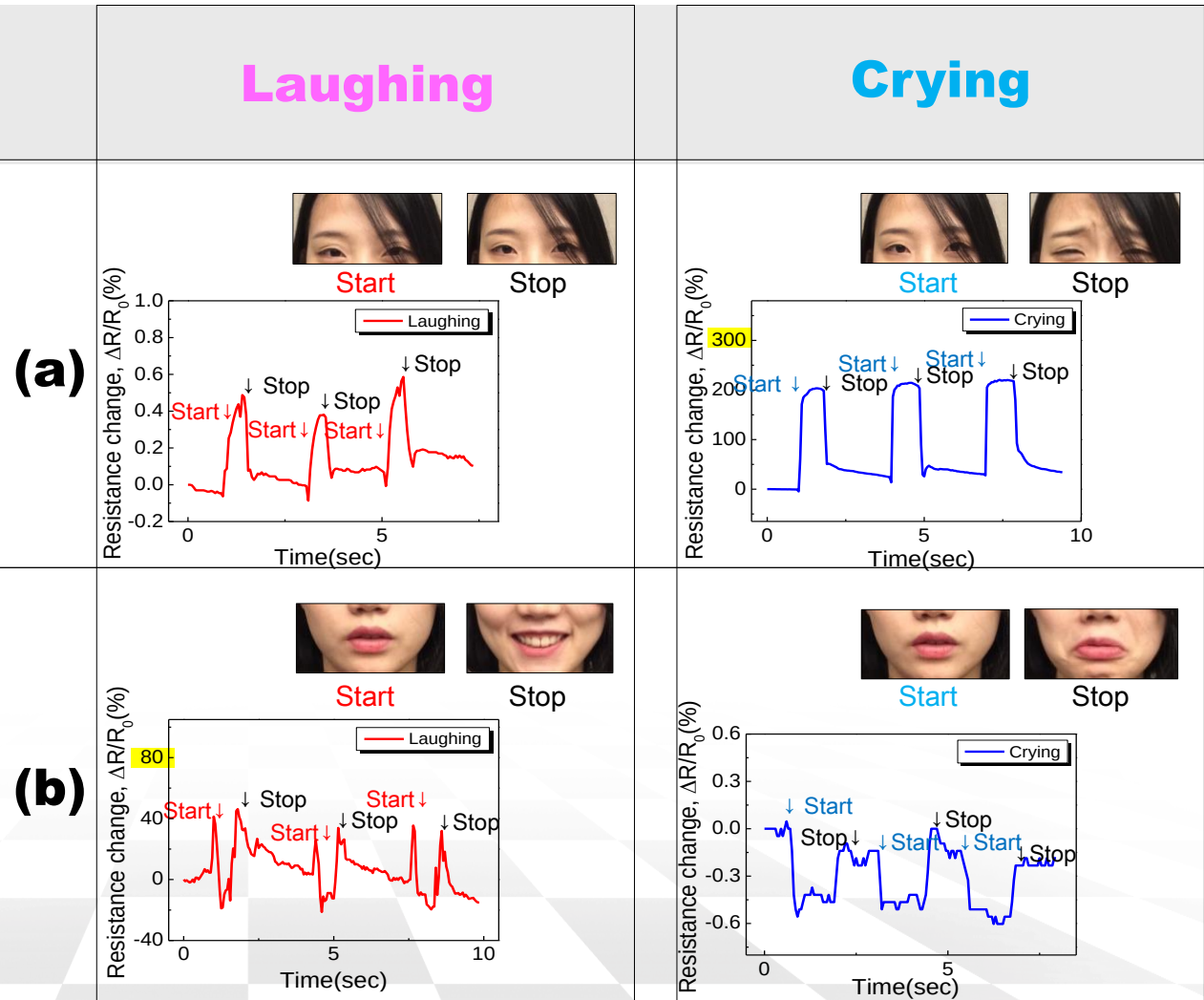
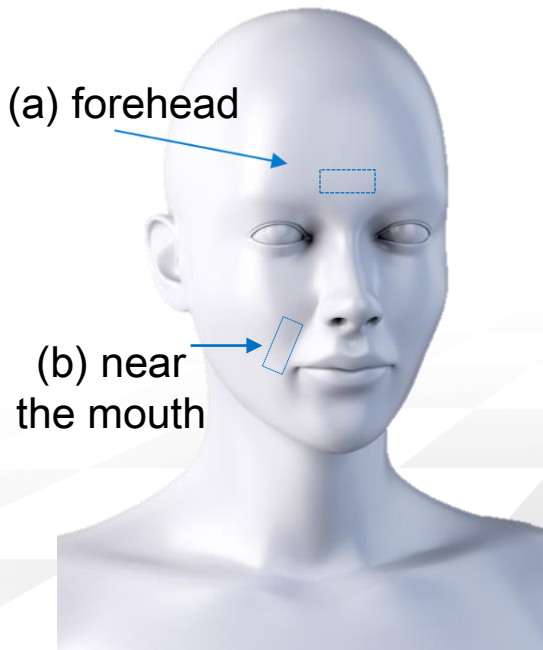
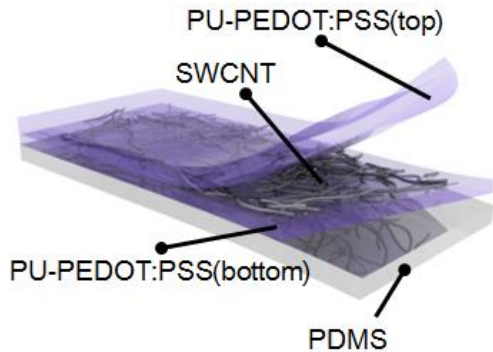
piezoelectric
pyroelectric
piezoresistive
chemresistive
thermorestistive
photoresponsive
electroactive

Stretchable

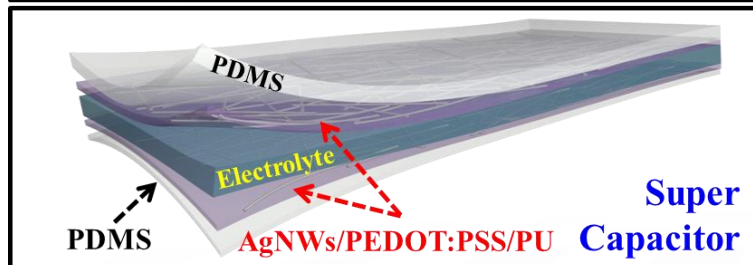
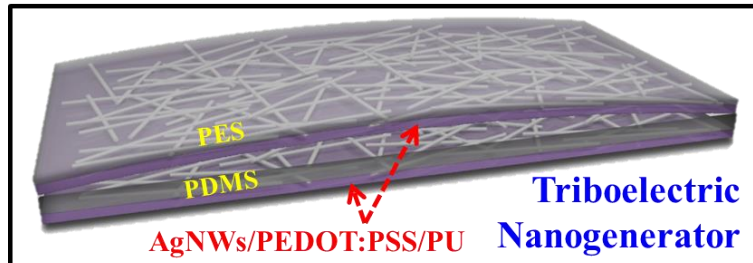
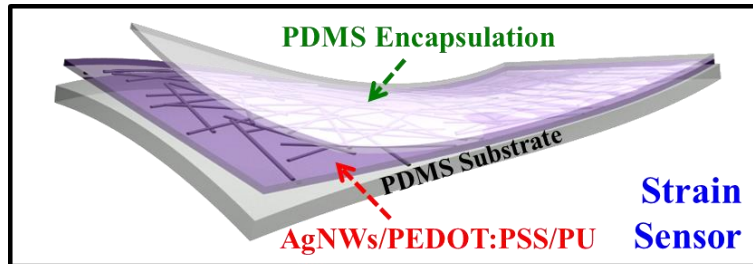
Stretchable, transparent and ultrasensitive strain sensor for emotion detection



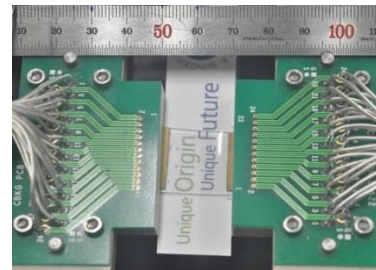
Stretchable, transparent and ultrasensitive strain sensor for emotion detection



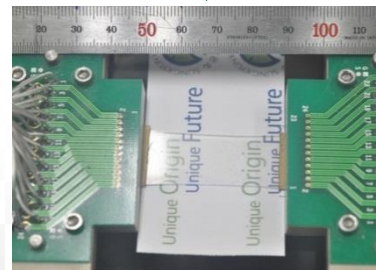
Stretchable, transparent, ultrasensitive, self-powered strain sensor for activity monitoring



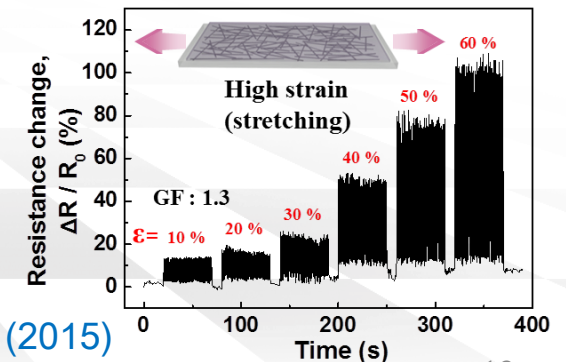
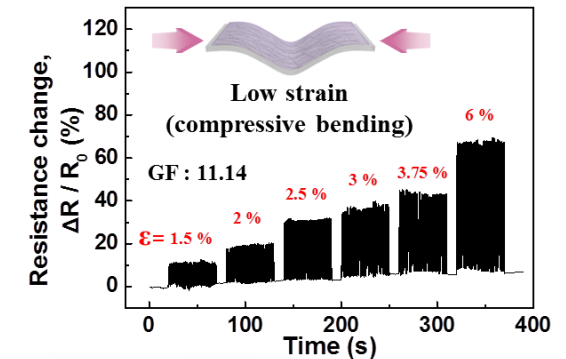
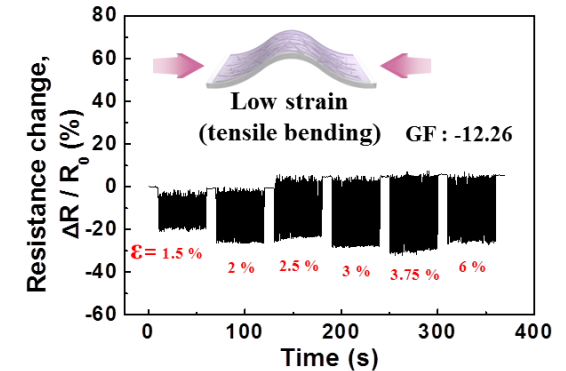
Evaluation



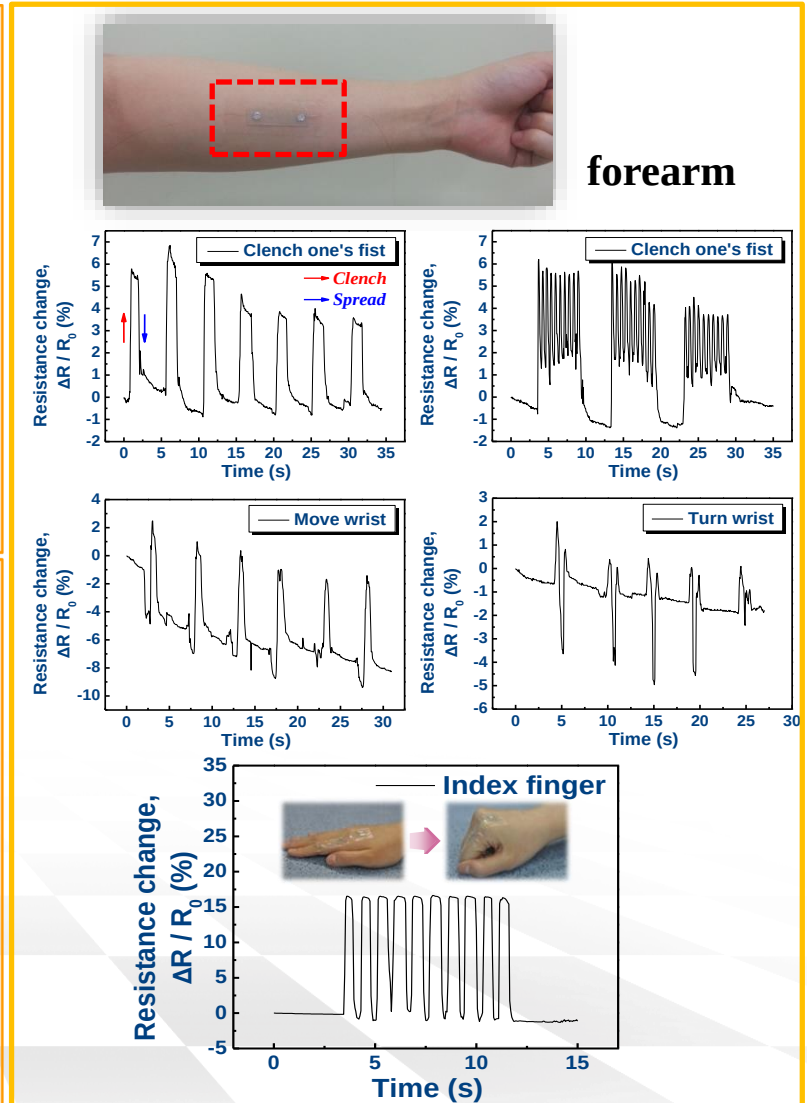
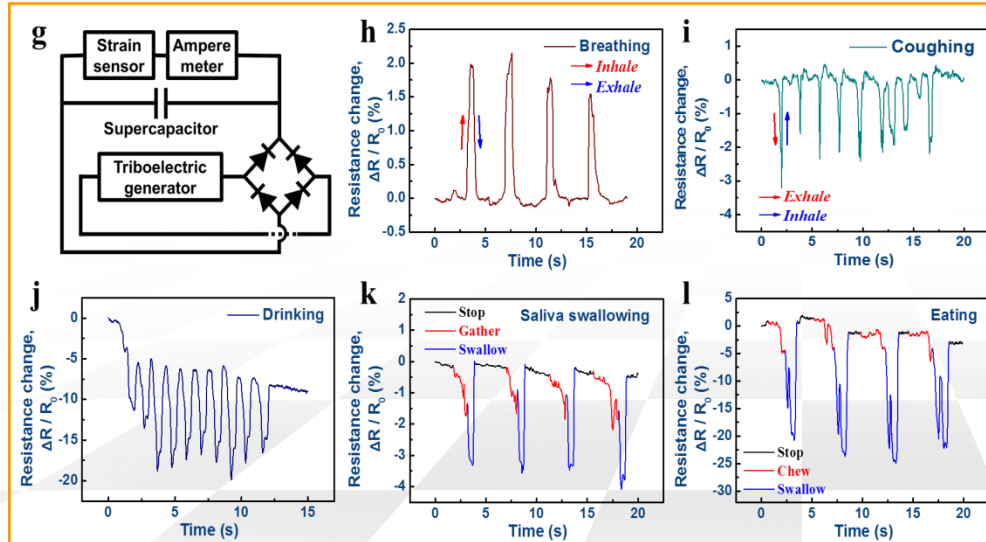
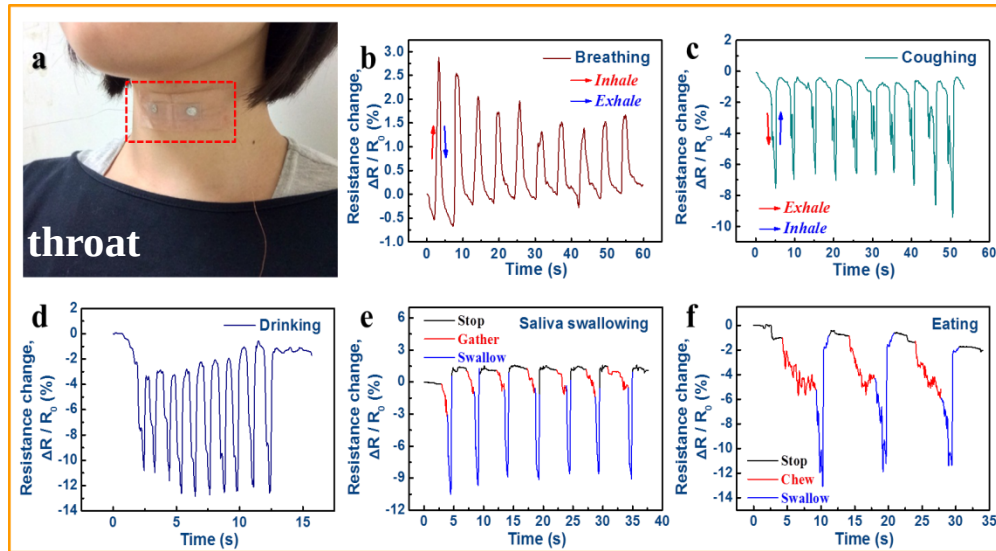
Strain : 0 %



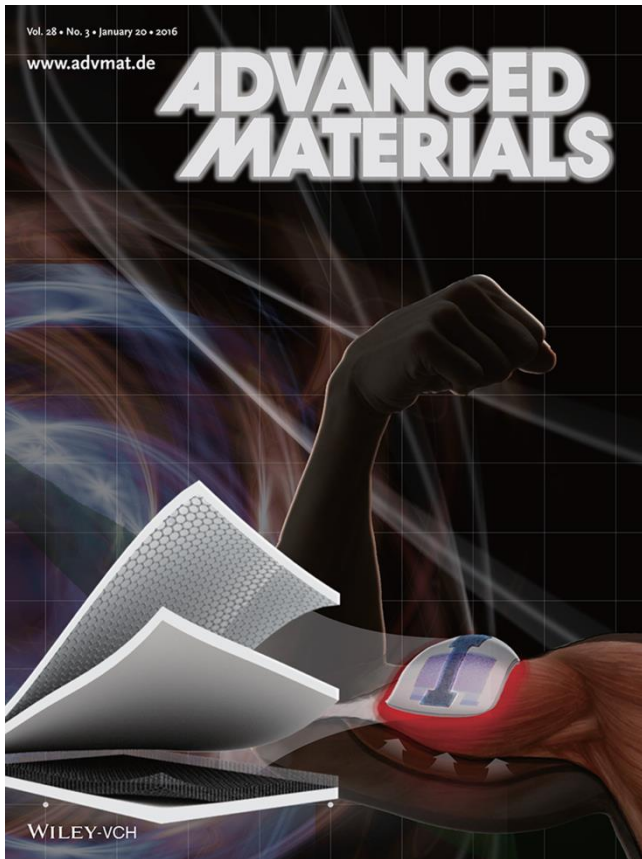
Strain : 100 %



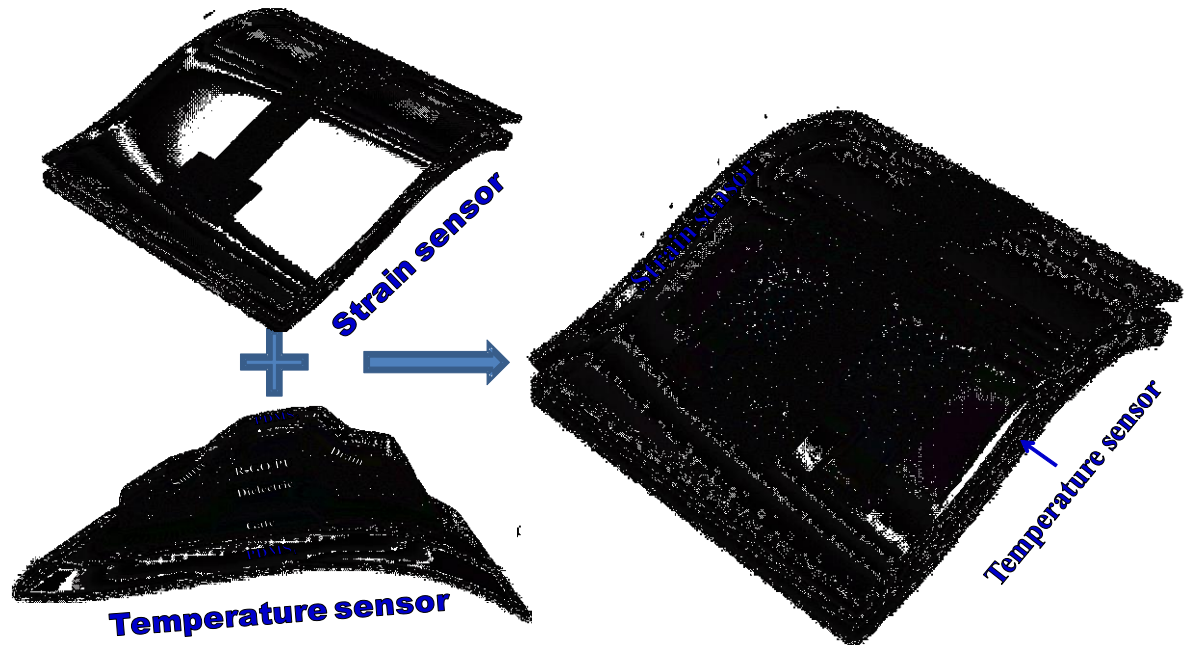
Stretchable, transparent, ultrasensitive, self-powered strain sensor for activity monitoring



All-elastomeric transparent and stretchable multi-sensors for activity monitoring



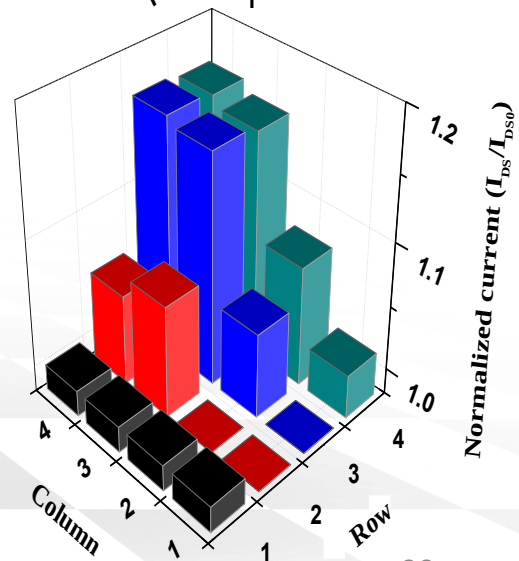
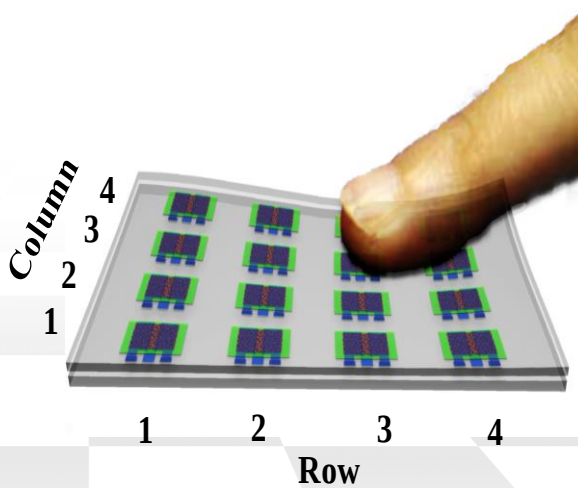
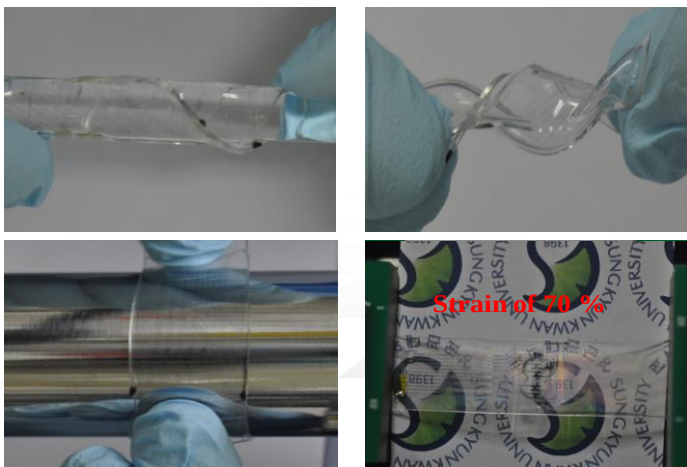
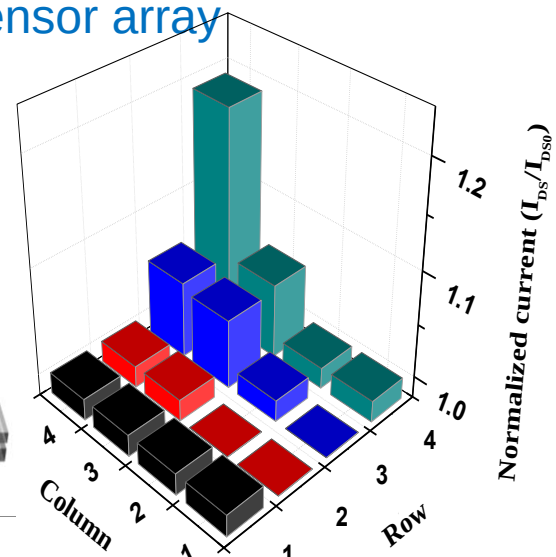
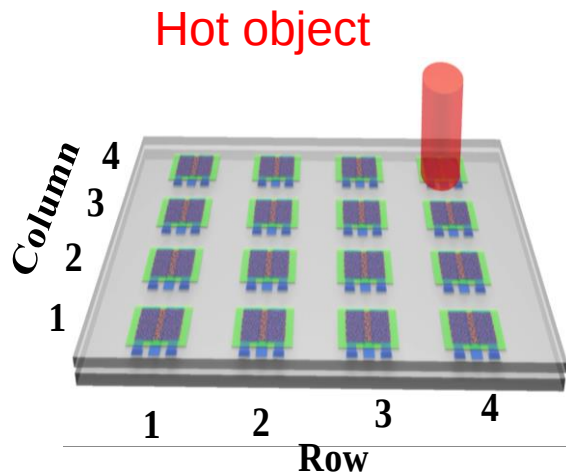
AgNWs/PEDOT:PSS-PU
resistor



Reduced graphene oxide-PU channel,
PU gate dielectric and
PEDOT:PSS-PU electrode

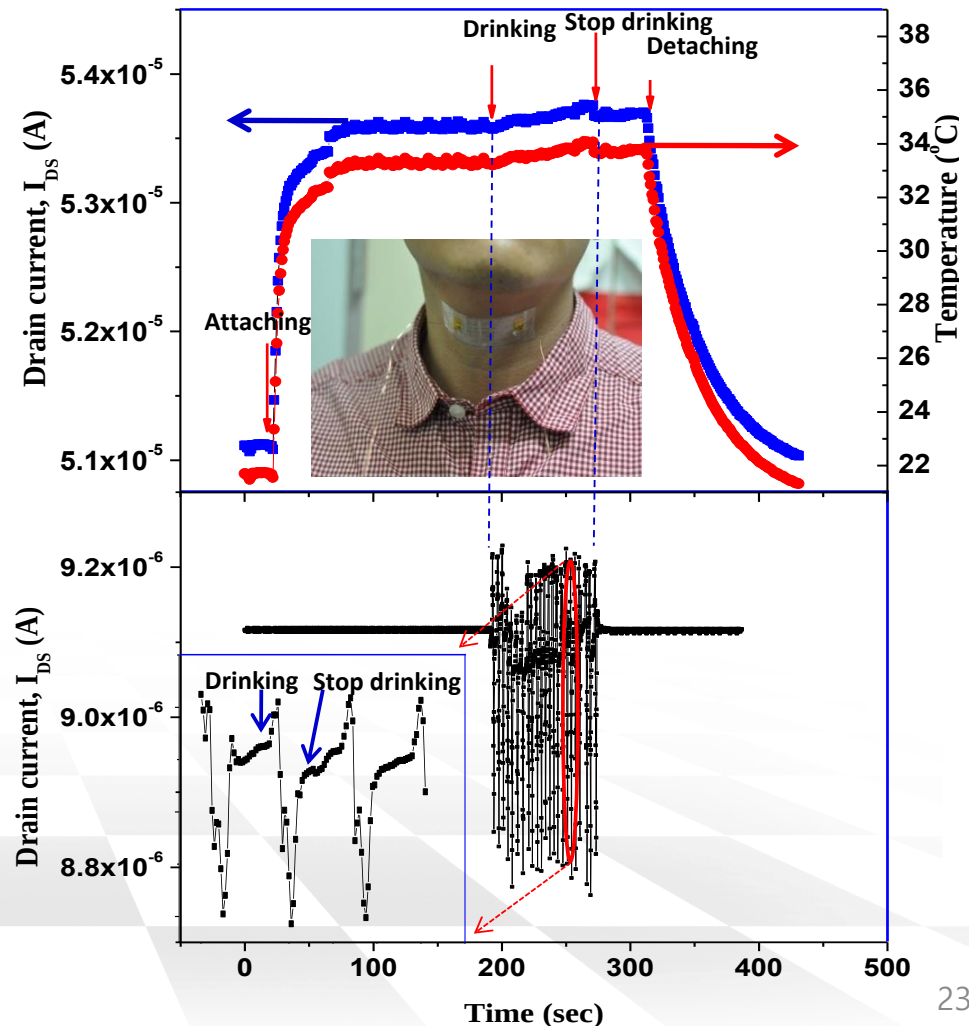
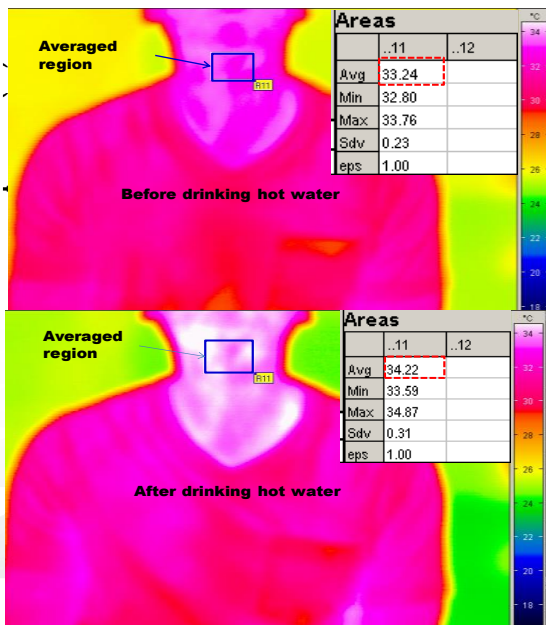
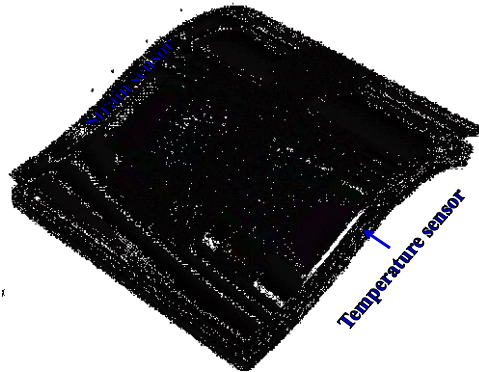
All-elastomeric transparent and stretchable multi-sensors for activity monitoring

Monitoring thermal distribution by FET temperature sensor array



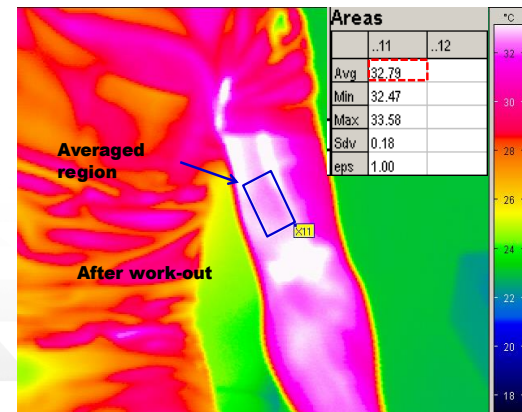
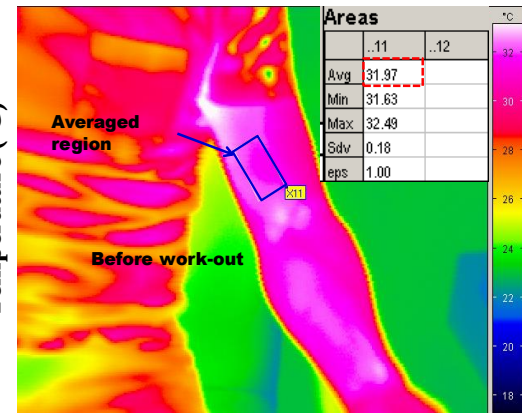
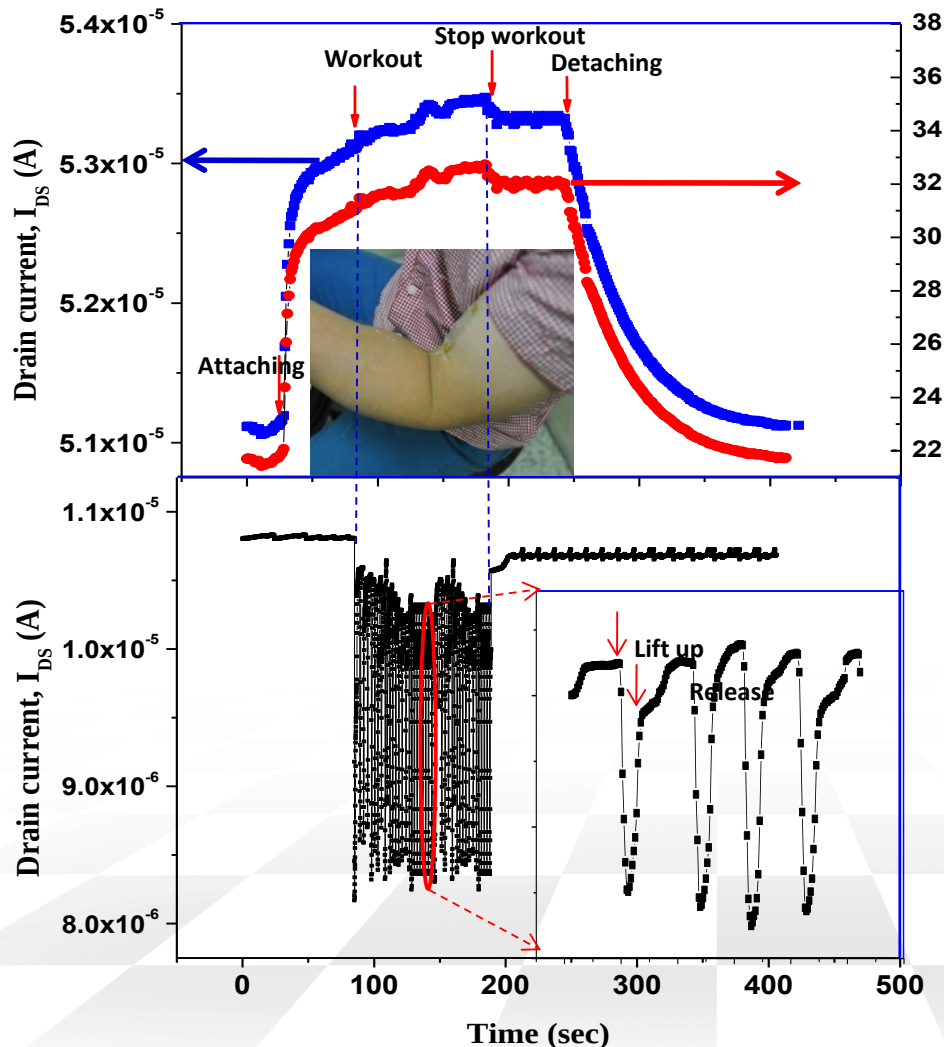
All-elastomeric transparent and stretchable multi-sensors for activity monitoring

Simultaneous monitoring skin temperature and muscle movement during drinking hot water

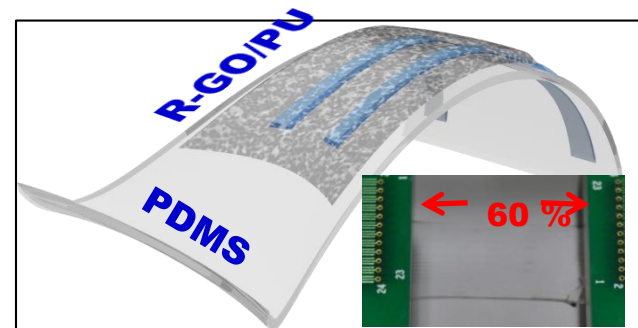
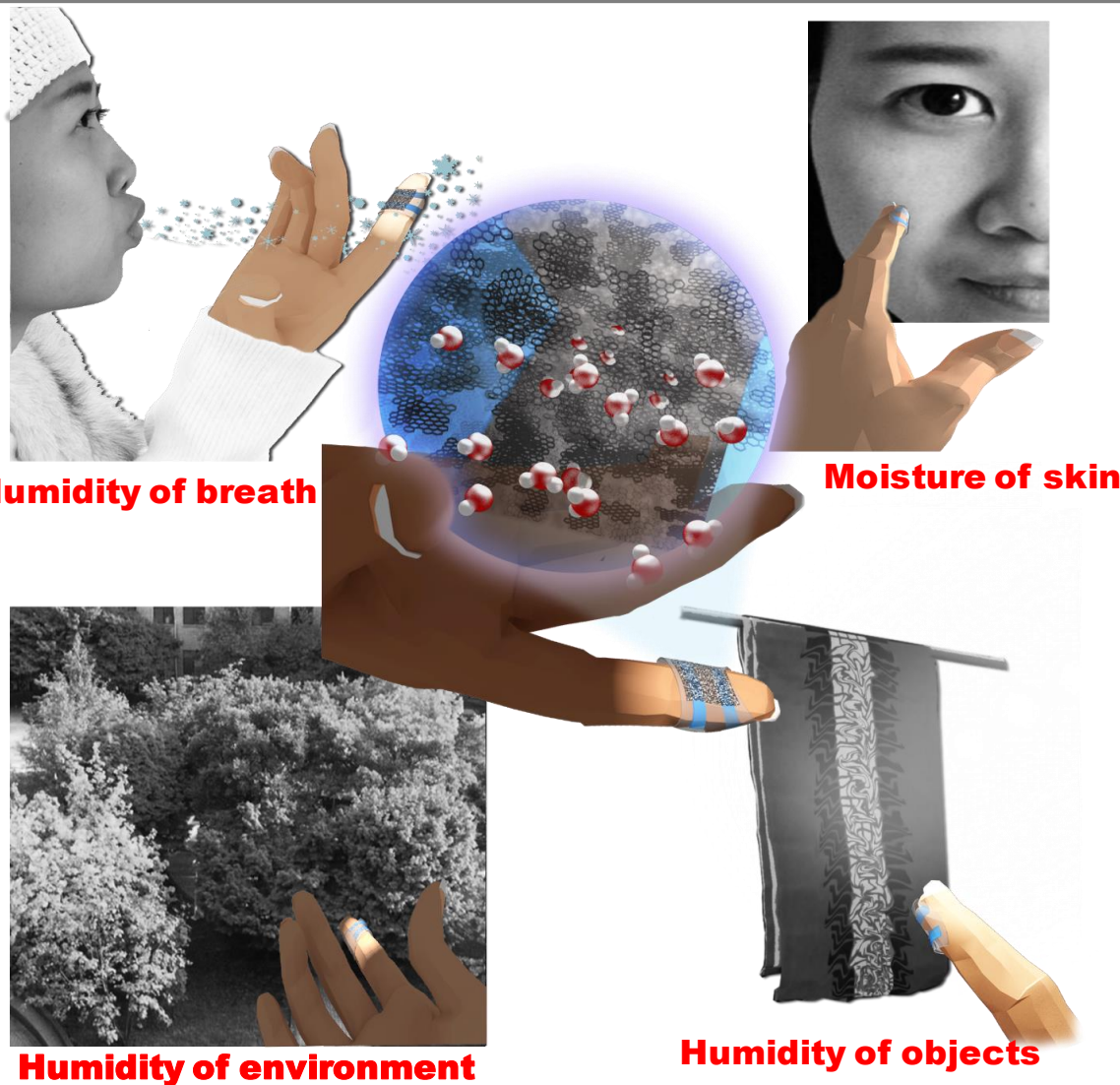


All-elastomeric transparent and stretchable multi-sensors for activity monitoring

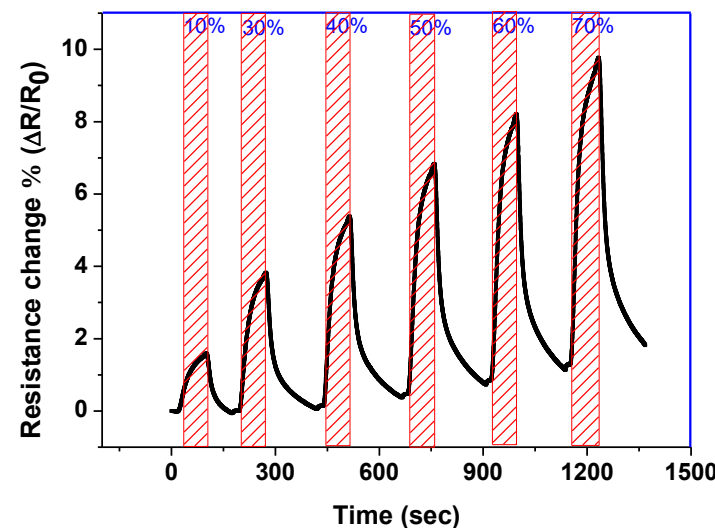
Simultaneous monitoring skin temperature and muscle movement during workout



Stretchable transparent humidity sensor for hydration monitoring



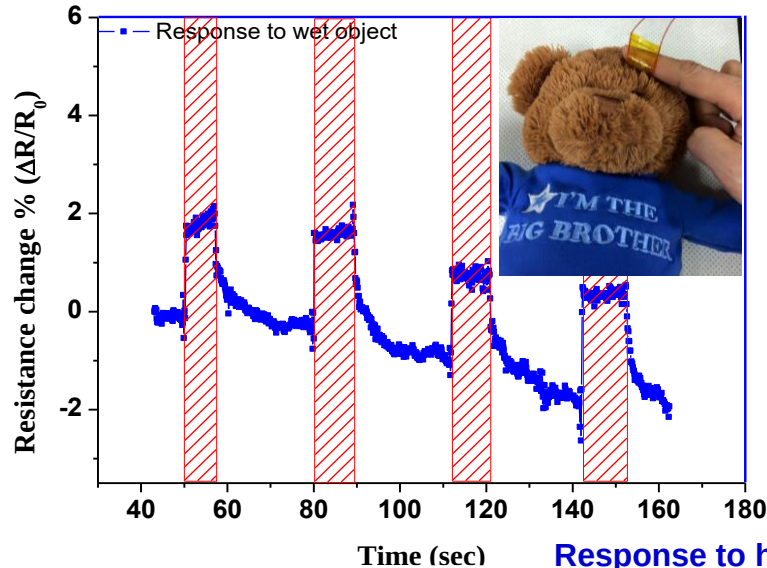
Response to humidity (10 % - 70 %)



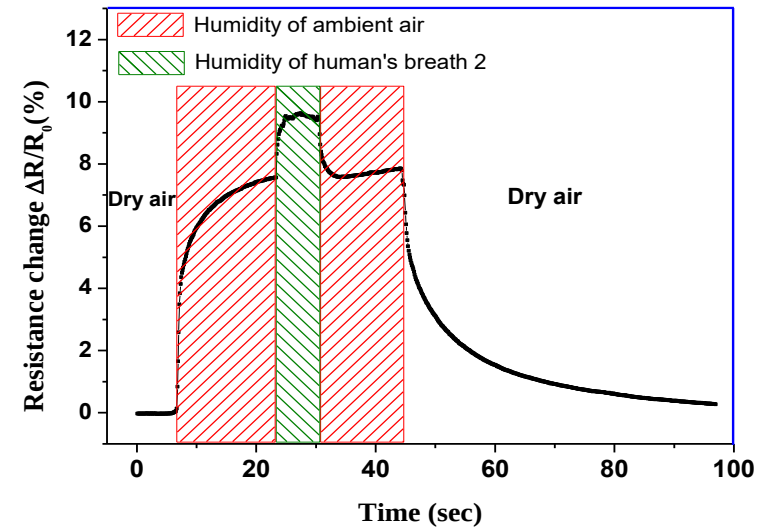
Stretchable transparent humidity sensor for hydration monitoring



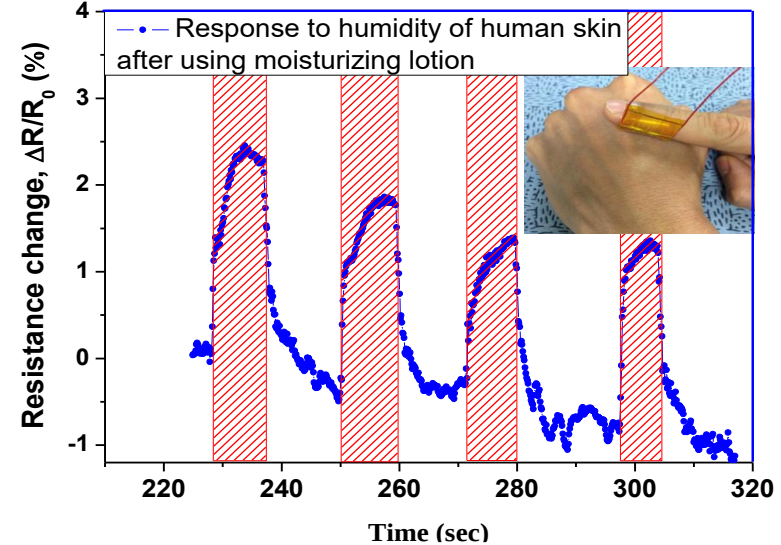
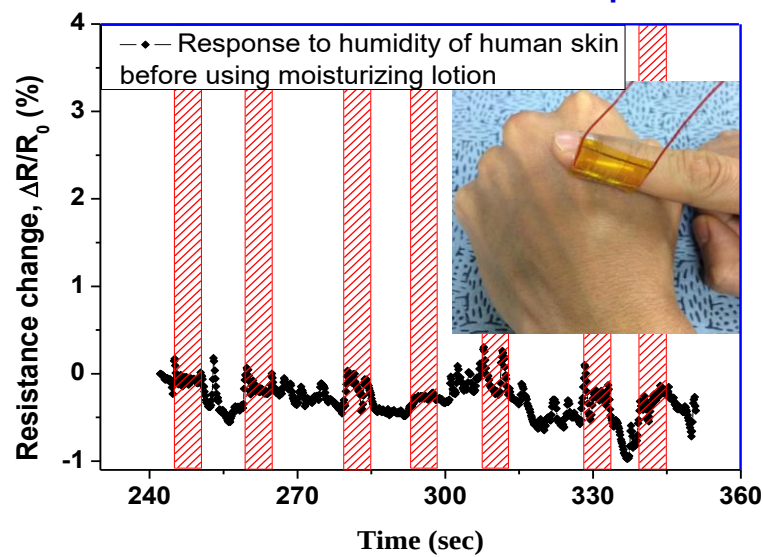
Response to humidity of wet object



Response to humidity of human breath

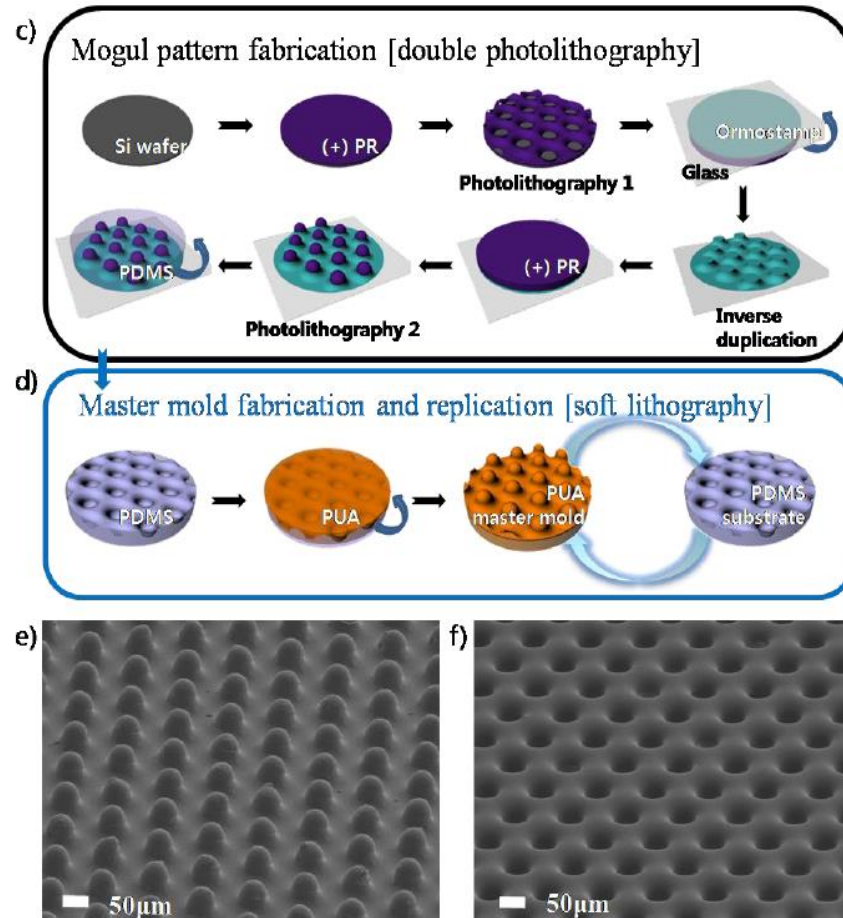
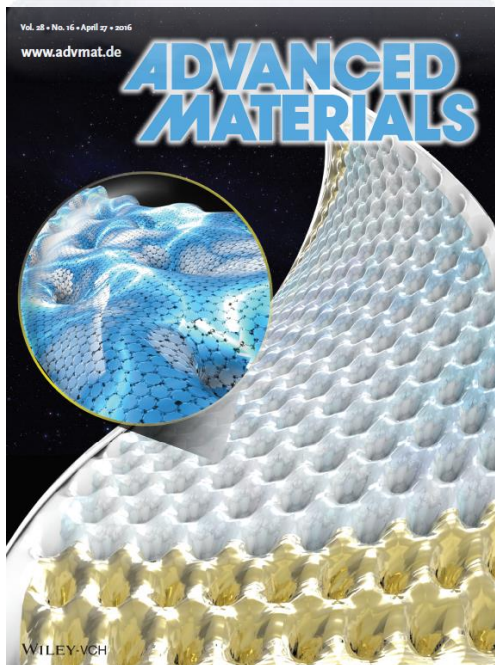


Response to hydration of human skin



Approach 2: Mogul-patterned elastomeric substrate

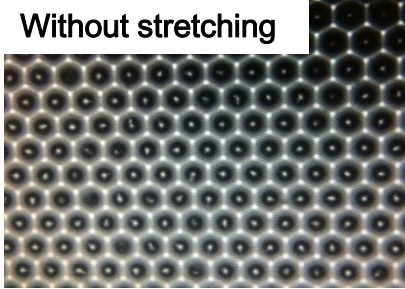
An omnidirectionally stretchable substrate for structural engineering



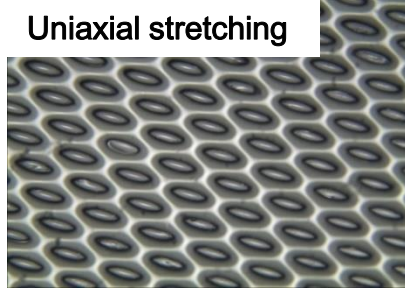
Approach 2: Mogul-patterned elastomeric substrate

Optical image depending on stretching direction

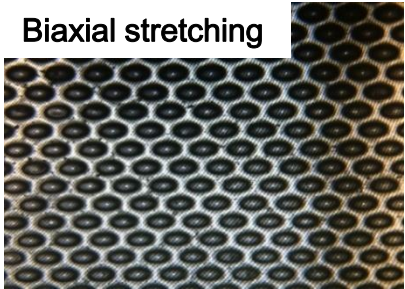
Without stretching



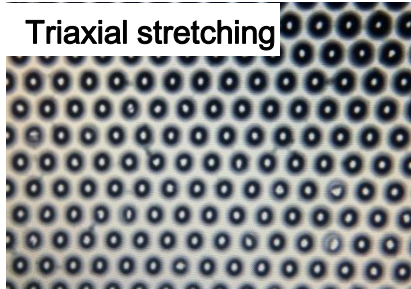
Uniaxial stretching



Biaxial stretching



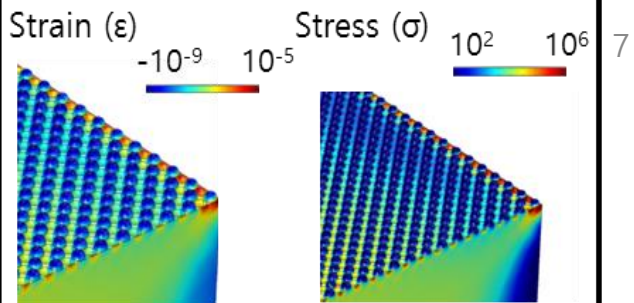
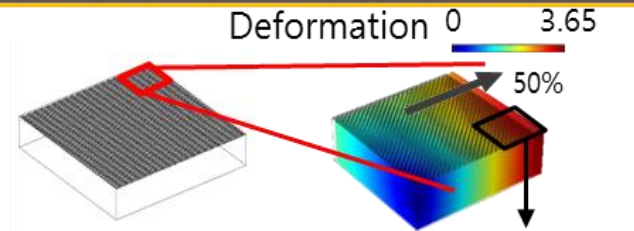
Triaxial stretching



Small tensile on bumps and valleys under elongation

Omniaxially stretchable

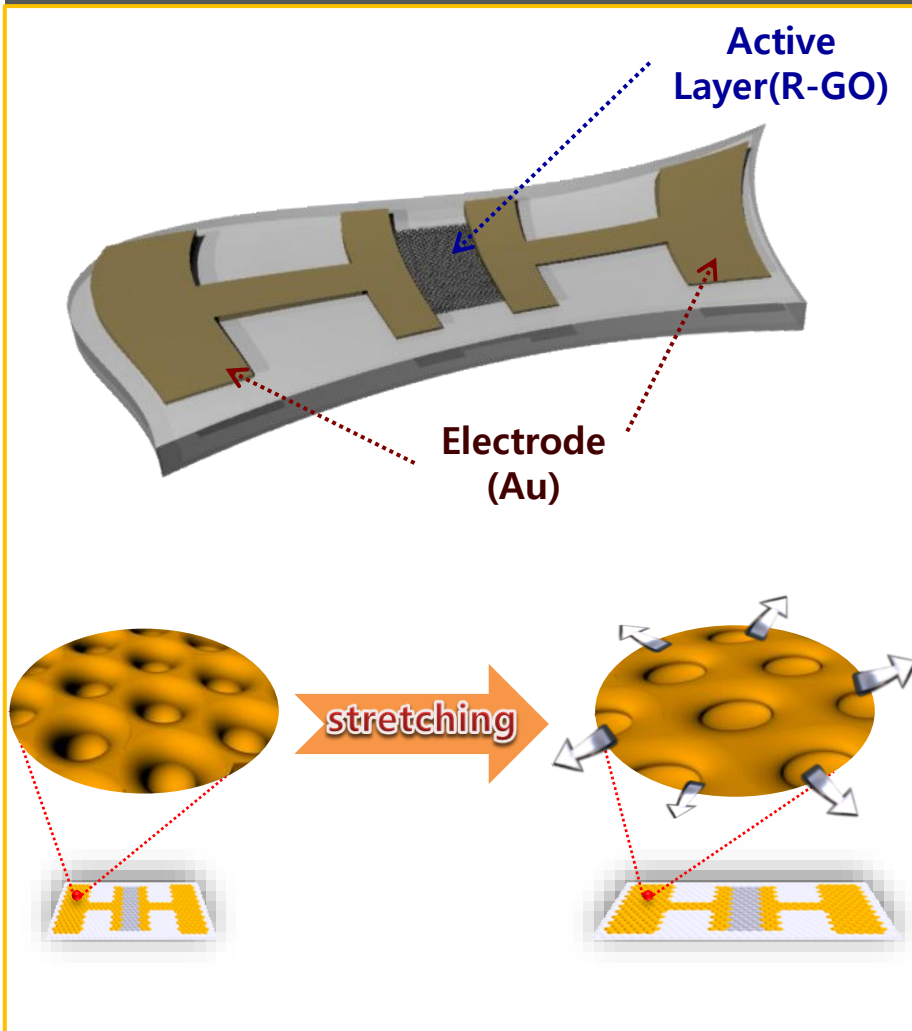
Computer simulation for deformation, strain, and stress



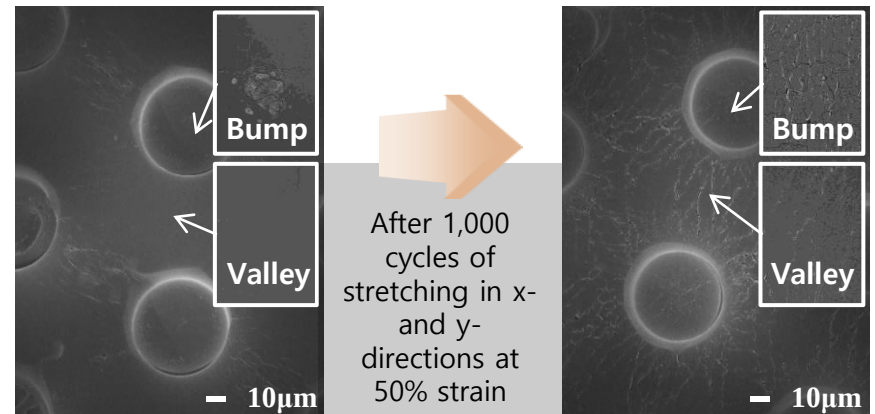
Omniaxially stretchable RGO gas sensor on mogul-patterned elastomeric substrate



Structure



Stability of Au electrode (70nm)



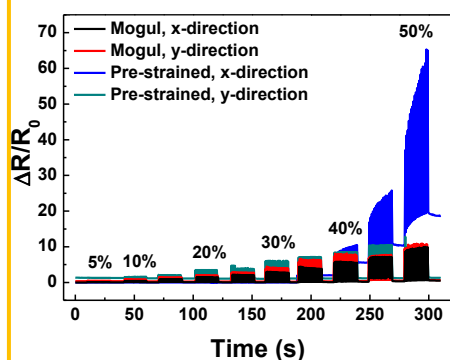
Omniaxially stretchable RGO gas sensor on mogul-patterned elastomeric substrate



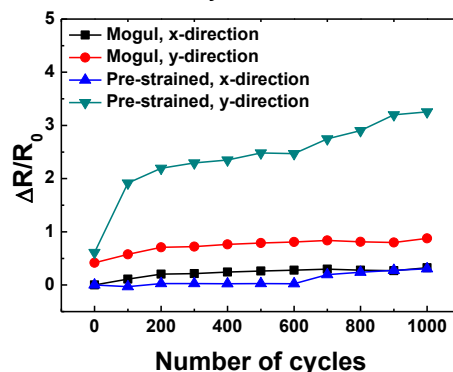
Investigation of electrical stability

Electrical stability of Au electrode

[Dynamic test]

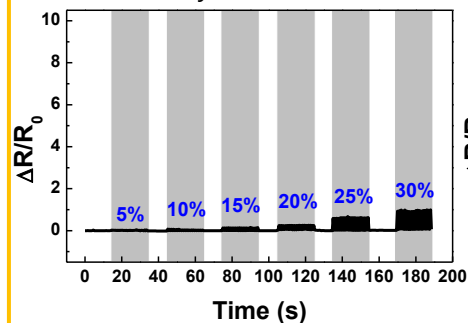


[Cyclic test]

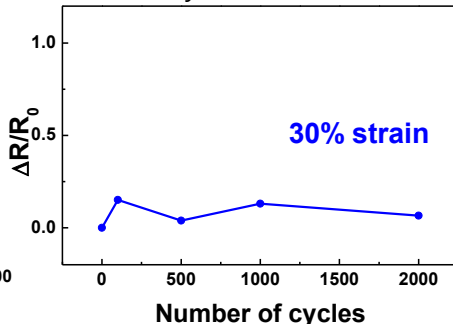


Electrical stability of chemical sensor

[Dynamic test]

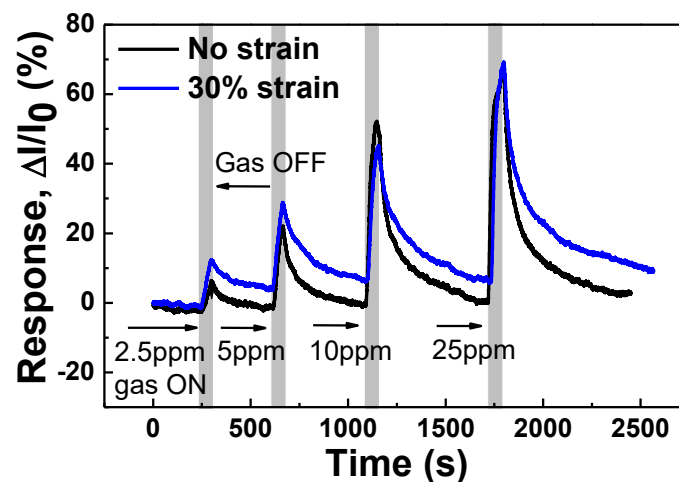


[Cyclic test]



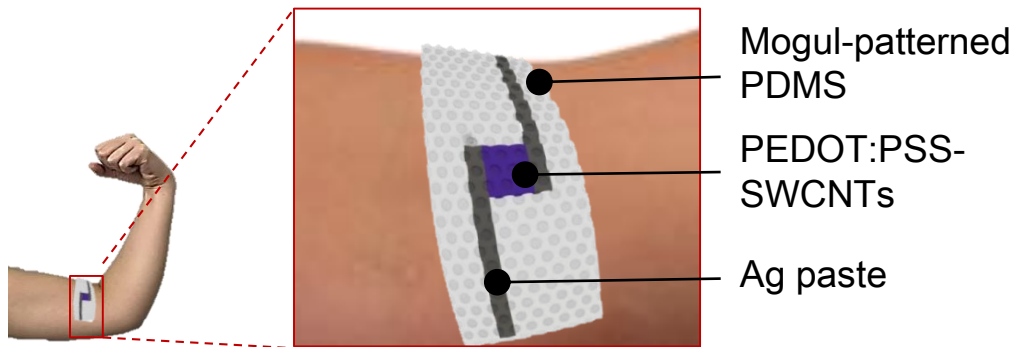
NO₂ sensing

Under stretched condition



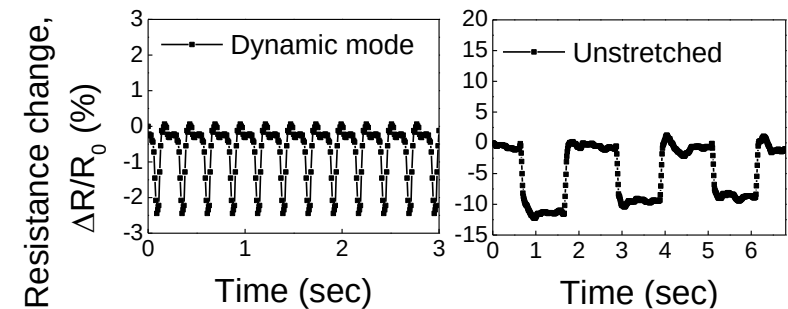
Omaniaxially stretchable piezoresistive pressure sensor on mogul-patterned substrate

Device Structure



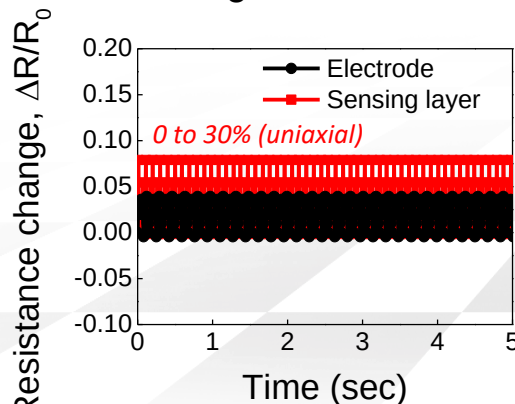
Pressure responsivity

- ✓ Unstretched state

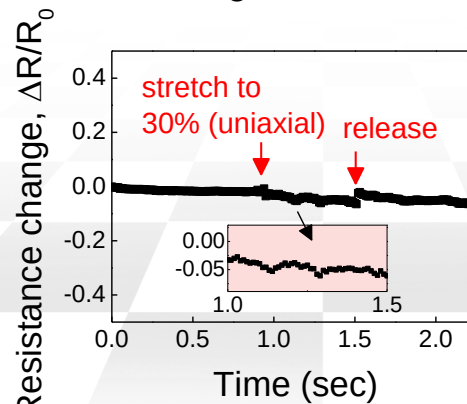


Stability under stretching

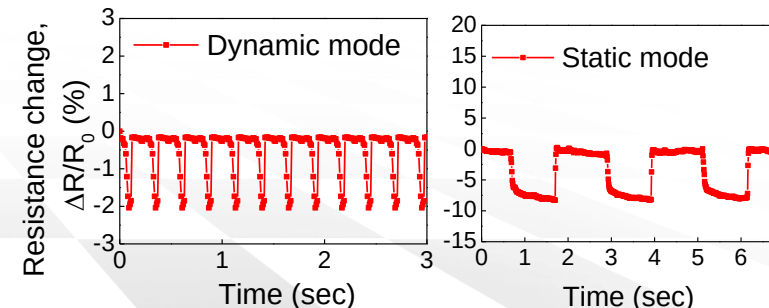
- ✓ Materials stability under stretching



- ✓ Device stability under stretching



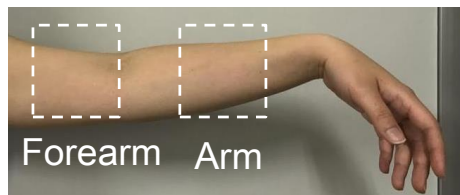
- ✓ Under 30% stretching state



Omniaxially stretchable piezoresistive pressure sensor on mogul-patterned substrate

Demonstration : Tremor detection

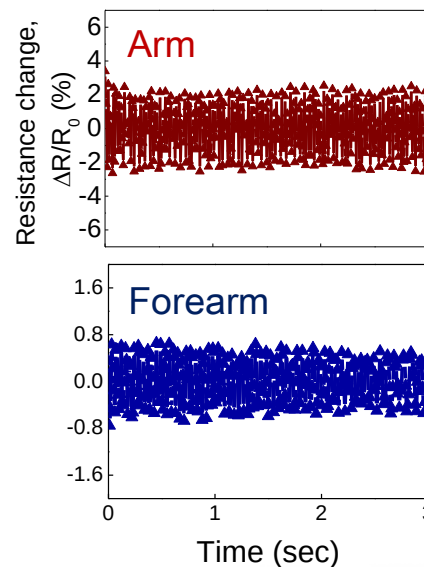
- ✓ Skin area



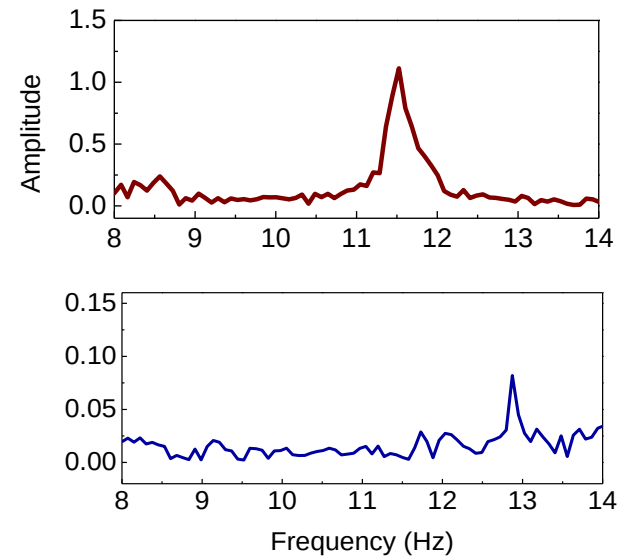
- ✓ Demonstration setup



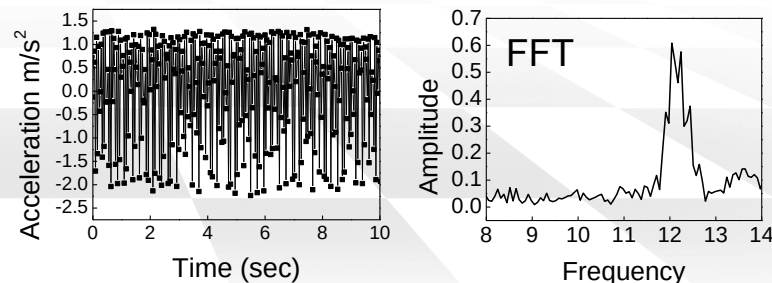
- ✓ Vibration detection using the device



- ✓ FFT (fast Fourier transform)

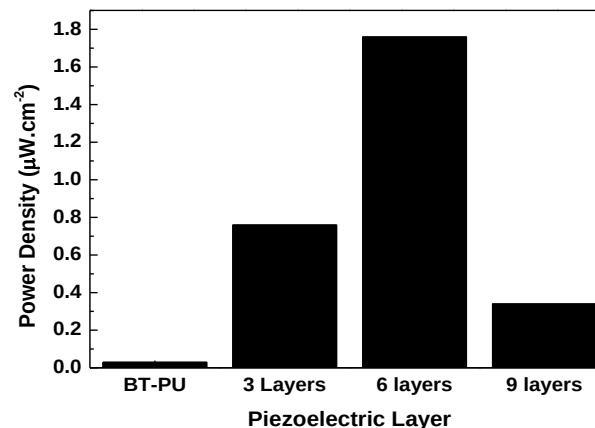
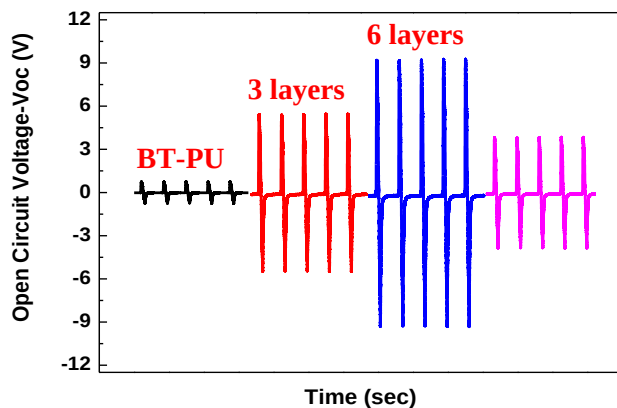
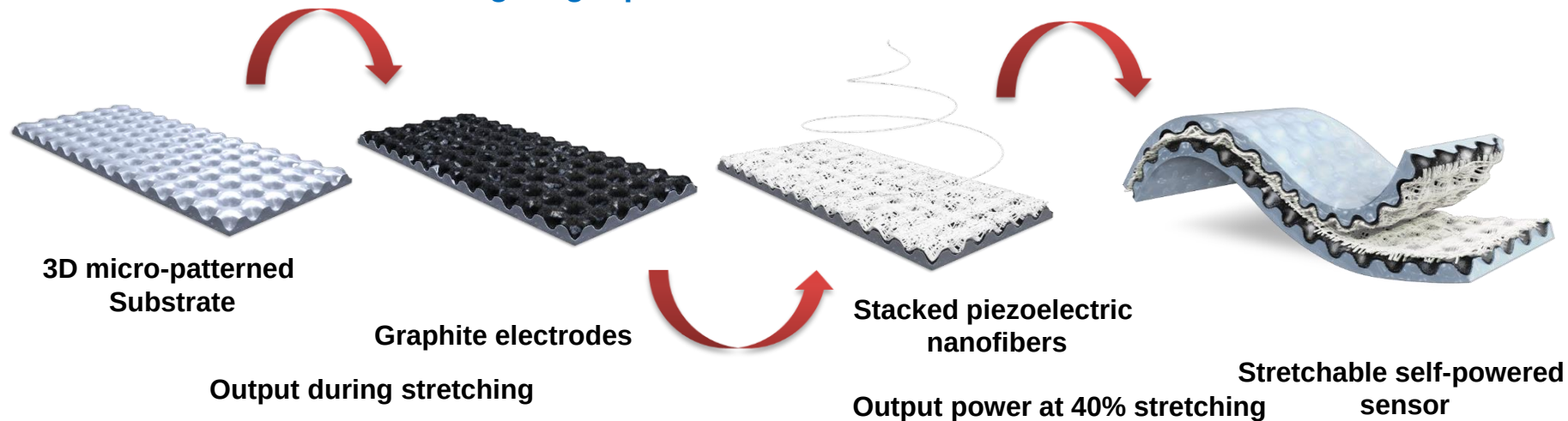


- ✓ Vibration detection using the accelerometer in smartphone

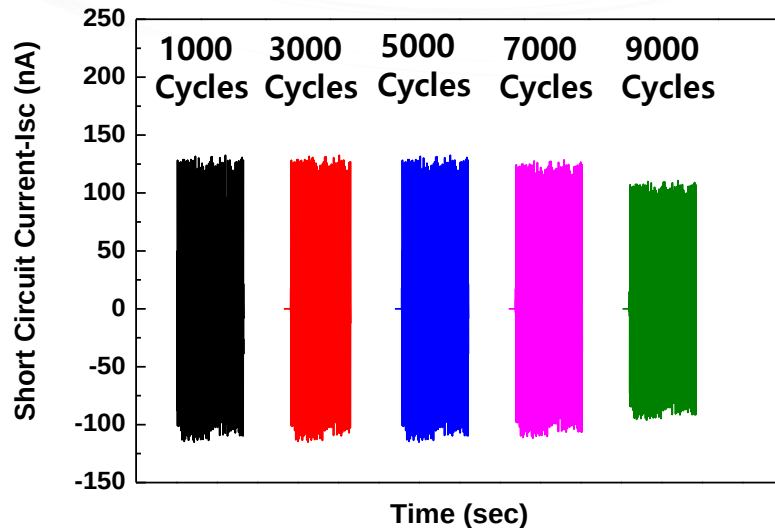
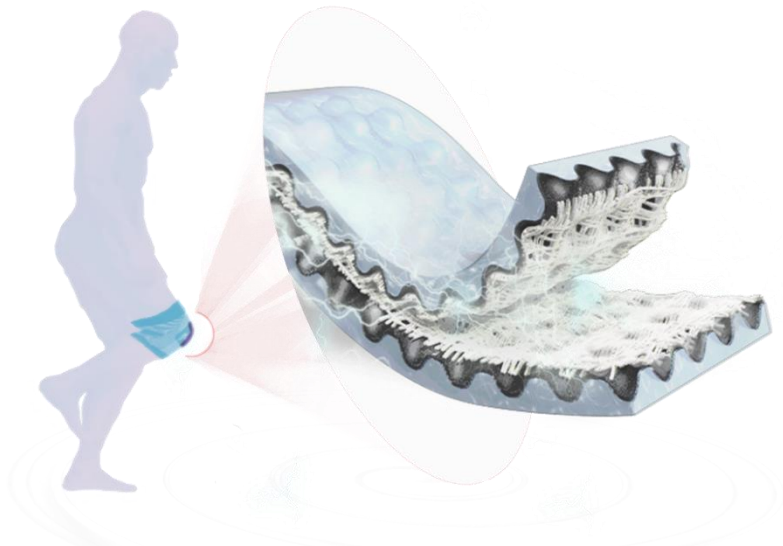


Omniaxially stretchable self-powered piezoelectric device

Stacked mat of BT NPs-PU nanocomposite and P9VDF-TrFE) nanofibers on stress-relieving mogul-patterned elastomeric substrate

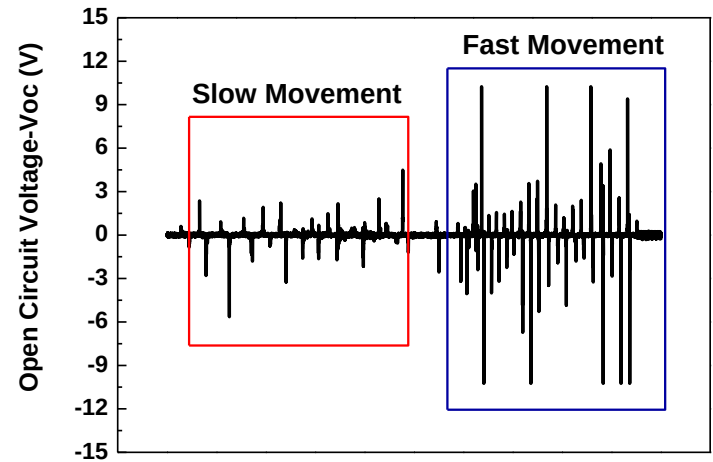


Omniaxially stretchable self-powered piezoelectric device

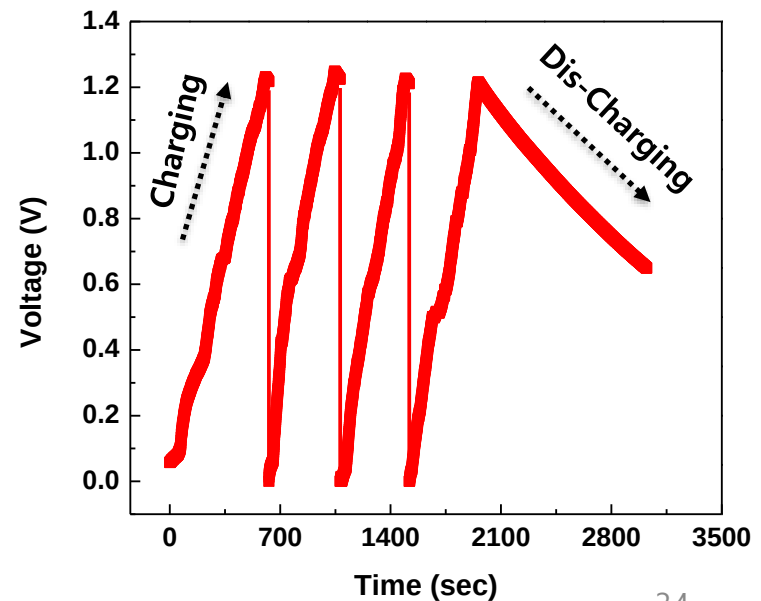


Stability at 30% stretching strain

Walking Pattern Detection



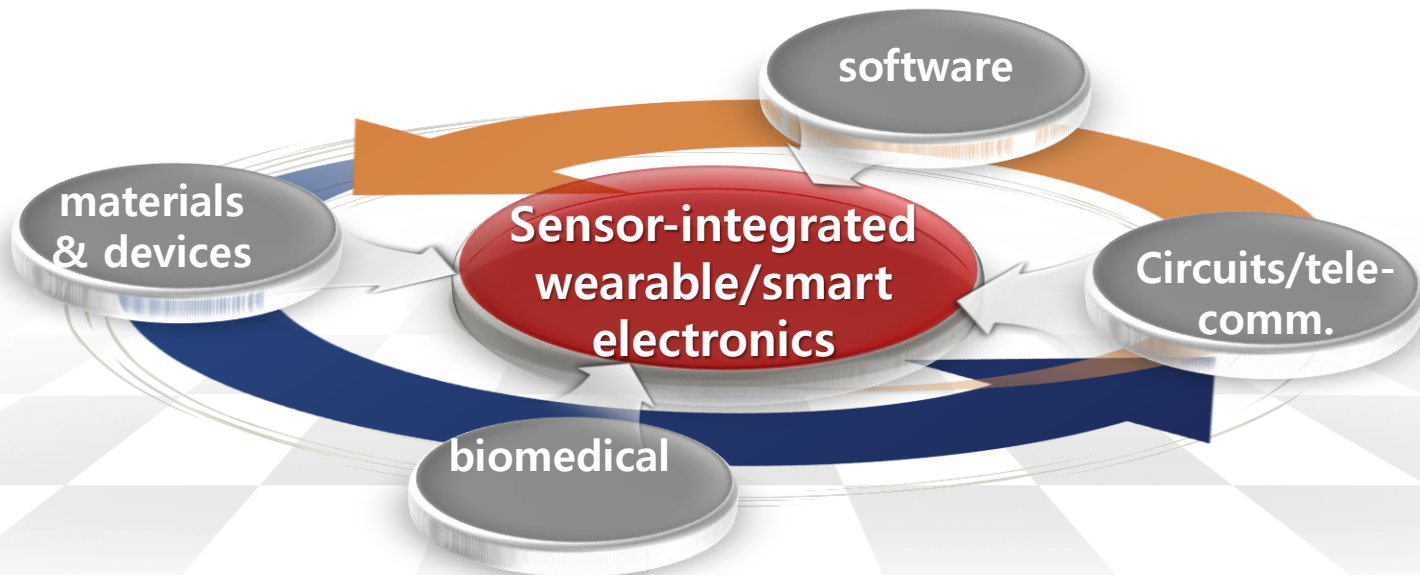
Charge Storage during walking



Perspectives



- Efforts toward the improvement of stability and reliability of the sensing nano-materials are required for real applications.
- Sensor-integrated systems by combining MCU, communication, energy and sample handling devices need to be developed by considering the specific service needs.
- Collaborative research is essential for success.



Thank you very much.

